# Competency Based Learning Material

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| Sector: **AGRICULTURE, FORESTRY AND FISHERY** | |
| Qualification : **ORGANIC AGRICULTURE PRODUCTION NC II** | |
| Unit of Competency: | **RAISE ORGANIC CHICKEN** |
| Module Title | **RAISING ORGANIC CHICKEN** |
| **MDM SAGAY COLLEGE INC.** | |

# ORGANIC AGRICULTURE PRODUCTION NC II

The **ORGANIC AGRICULTURE PRODUCTION NC II** Qualification consists of competencies that a person must achieve to produce organic farm products such as chicken and vegetables including producing of organic supplements such as fertilizer, concoctions and extracts. It has two (2) elective competencies which are on raising organic hogs and raising organic small ruminants.
This Qualification is packaged from the competency map of the Agri-Fishery Sector.

The units of competency comprising this qualification includes the following:

| CODE | BASIC COMPETENCIES |
|---|---|
| 500311105 | Participate in workplace communication |
| 500311106 | Work in a team environment |
| 500311107 | Practice career professionalism |
| 500311108 | Practice occupational health and safety procedures |

| CODE | COMMON COMPETENCIES |
|---|---|
| AFF 321201 | Apply safety measures in farm operations |
| AFF 321202 | Use farm tools and equipment |
| AFF 321203 | Perform estimation and calculations |
| TRS311201 | Develop and update industry knowledge |
| AGR321205 | Perform record keeping |

| CODE | CORE COMPETENCIES |
|---|---|
| AGR612301 | Raise organic chicken |
| AGR611306 | Produce organic vegetables |
| AGR611301 | Produce organic fertilizer |
| AGR611302 | Produce organic concoctions and extract |
| **CODE** | **ELECTIVE COMPETENCIES** |
| AGR612302 | Raise organic hogs |
| AGR612303 | Raise organic small ruminants |

# HOW TO USE THIS MODULE

Welcome to the Module on Raising Organic Chicken. This module contains training materials and activities for you to complete.

The unit of competency on *Raise Organic Chicken* contains knowledge, skills and attitudes required for an Organic Agriculture Production NC II course.

You are required to go through a series of learning activities in order to complete each of the learning outcomes of the module. In each learning outcome there are **Information Sheets, Operation Sheets**, **Job Sheets** and **Task Sheets**. Follow these activities on your own and answer the Self-Check at the end of each learning activity.

If you have questions, don't hesitate to ask your trainer for assistance.

**Recognition of Prior Learning (RPL)**

You may already have some of the knowledge and skills covered in this module because you have:

- been working for some time
- already have completed training in this area.

If you can demonstrate to your trainer/instructor that you are competent in a particular skill or skills, talk to him/her about having them formally recognized so you don't have to do the same training again. If you have a qualification or Certificate of Competency from previous trainings show it to your trainer. If the skills you acquired are still current and relevant to this module, they may become part of the evidence you can present for RPL. If you are not sure about the currency of your skills, discuss it with your trainer.

After completing this module ask your trainer to assess your competency. Result of your assessment will be recorded in your competency profile. All the learning activities are designed for you to complete at your own pace.

Inside this module you will find the activities for you to complete followed by relevant information sheets for each learning outcome. Each learning outcome may have more than one learning activity.

# MODULE CONTENT

Program/Course : Organic Agriculture Production NC II
Unit of Competency : Raise Organic Chicken
Module : Raising Organic Chicken

**INTRODUCTION:**

This module contains information and suggested learning activities on raising organic chicken. It includes activities and materials on Organic Agriculture Production NC II

Completion of this module will help you better understand the succeeding module on the raising organic chicken.

This module consists of 4 learning outcomes. Each learning outcome contains learning activities supported by each instruction sheets. Before you perform the instructions, read the information sheets and answer the self-check and activities provided to ascertain to yourself and your trainer that you have acquired the knowledge necessary to perform the skill portion of the particular learning outcome.

Upon completion of this module, report to your trainer for assessment to check your achievement of knowledge and skills requirement of this module. If you pass the assessment, you will be given a certificate of completion.

## SUMMARY OF LEARNING OUTCOMES:

Upon completion of the module you should be able to:

LO1 Select healthy stocks and suitable housing
LO2 Set-up cage equipment
LO3 Feed chicken
LO4 Grow and harvest chicken

**Learning Outcome #1 Select healthy stocks and suitable housing**

**Assessment Criteria:**

Breed/strains breeds are identified as per PNS-Organic Agriculture-Livestock and GAHP Guidelines

Healthy chicks are selected based on industry acceptable indicator for healthy chicks

Suitable site for chicken house are determined based on PNS recommendations.

Chicken house design is prepared based PNS recommendations.

House equipment installation design is prepared in line with PNS recommendation and actual scenario.

**Contents:** Chicken breeds Identification
Healthy chick's selection
Determining suitable site for chicken house
Chicken house design preparation
House equipment installation

**Tools, Materials and Equipment and Facilities**

Farm
Feeding troughs
Waterers
Containers of concoction
Chicken/ chicks
Rice hull
Saw dust
Coco coir
Rice straw
PPE ( Boots, surgical masks, disposable gloves, overall)

**References:**

https://starmilling.com/poultry-chicken-breeds

https://www.google.com/search?sxsrf=AOaemvKfBotEpb8oaDfLzaZm4Le020JS1w:1631077568545&q=suitable+site+for+chicken+house&spell=1&sa=X&ved=2ahUKEwjGuMu4ze7yAhXNAogKHQSLCcoQBSgAegQIAhAx&biw=1367&bih=630

https://the-chicken-chick.com/tips-for-choosing-healthy-chicks/

https://poultrymanual.com/philippines-chicken-house-design

# LEARNING EXPERIENCES/ACTIVITIES

## LO # 1 SELECT HEALTHY STOCKS AND SUITABLE HOUSING

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 1.1-1 on Chicken breeds identification | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.1-1 on Chicken breeds identification | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.1-1 on Chicken breeds identification | |
| Perform Job Sheet 1.1-1 on Chicken breeds identification | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 1.1-1 on Chicken breeds identification | |
| Read the attached Information Sheet 1.1-2 on Healthy chick's selection | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.1-2 on Healthy chick's selection | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.1-2 on Healthy chick's selection | |
| Perform Job Sheet 1.1-2 on Healthy chick's selection | Observe the demonstration of your trainer and check the procedures while performing the operation. |

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| Evaluate performance using performance checklist on 1.1-2 on Healthy chick's selection | |
| Read the attached Information Sheet 1.1-3 on Determining suitable site for chicken house | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.1-3 on Determining suitable site for chicken | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.1-3 on Determining suitable site for chicken house | |
| Read the attached Information Sheet 1.1-4 on Chicken house design preparation | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.1-4 on Chicken house design preparation | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.1-4 on Chicken house design preparation | |

***After doing this activity, you are now ready to proceed to the next module on Set-up cage equipment.***

# INFORMATION SHEET 1.1-1

## Chicken breeds identification

**Learning Objective:** After reading this information sheet, you should be able to identify chicken breeds.

**Introduction:**

**Australorp**

Around the same time that Orpingtons were being developed as a breed, Australorps were as well. Australians liked the black Orpingtons that were being brought over from England, and valued them for their egg laying ability. With maximum egg production in mind, Australians continued to develop their own districnt breed. The breed went by many names, struggling to distinguish itself from Orpingtons, and finally settled on Australorp in the 1920s.

These birds are known for their excellent egg production. You'll easily get 250 light brown eggs per year. The record holding hen laid 364 eggs in a 365 day period, without assistance of artificial lighting!

**Bielefelder**

The Bielefelder is a modern breed, developed in the early 1970s in Bielefeld, Germany. Poultry breeder Gerd Roth used genetics from a number of breeds including the Cuckoo Malines, Amrock, Wyandotte, and New Hampshire.

The Bielefelder managed to retain the best qualities of all these breeds. Consider this breed another example of extraordinary German engineering. They check off all the boxes on your “perfect chicken” wish list.

This dual-purpose breed is autosexing so males and females can be identified immediately upon hatching. Females have a chipmunk strip on their backs, while males are lighter in color and have a yellow spot on their heads. They mature to have a complex feather pattern which is best described as cuckoo red partridge.

Birds are very friendly and seek human interaction. They have a large frame that holds plenty of meat. Roosters can weigh 10 – 12 pounds! Their size and camouflaging feather pattern makes them perfect for free-range conditions.

Hens can produce upwards of 230 large eggs per year. Their eggs are a unique shade of brown with pink undertones that you won’t find anywhere else.

**Black Star / Red Star**

Red Stars and Black Stars are hybrids that have been bred to have their color at hatching linked to the sex of the chicken (pullet or cockerel). This makes chick sexing an easier process, and you as the purchaser are less likely to be surprised! Ever had a pullet start to crow one day? Not with these birds!

They've also been developed to be extremely good egg layers. Don't be surprised if you see 300 eggs in a year! Egg color and size will vary, depending on the cross-breeding.

Black Stars are a cross between a Rhode Island Red rooster and a Barred Rock hen. Red Star's are a cross between a Rhode Island Red rooster and either a White Rock, Silver Laced Wyandotte, Rhode Island White or Delaware hen.

**Java**

The Java is the 2nd oldest chicken breed developed in America, going back to 1835. Its ancestors come from the island of Java in the Far East. Javas are an excellent breed for free-ranging homesteads and are known for their egg production and table qualities. Javas come in white, black, mottled, and auburn. The Black Java is known for the brilliant beetle-green sheen of its feathers.

**Jersey Giant**

The Jersey Giant chicken was developed between 1870 and 1890 in New Jersey. You can probably guess that these birds are pretty big! Roosters weight in at 13 pounds, and hens can easily grow up to 10 pounds! They are the largest purebred chicken breed.

They are known to be fairly good layers compared to other large breeds, and are good winter layers. Expect about 260 large brown eggs per year.

**Maran**

Marans originated in western France and were imported in the 1930s. There are 9 recognized colors: Cuckoo, Golden Cuckoo, Black, Birchen, Black Copper, Wheaten, Black-tailed Buff, White and Columbian. If you find Marans chicks at a feed store, they will most likely be the Black Copper or Cuckoo variety. Hens are active and enjoy free ranging, and also have friendly, outgoing personalities.

Marans are renowned for their dark chocolate brown eggs. If you're looking for unique eggs, these are quite the conversation peace! You'll get about 150-200 each year in your nesting boxes.

**Naked Neck**

The Naked Neck is a breed of chicken that is naturally devoid of feathers on its neck and vent. The breed is also called the Transylvanian Naked Neck, as well as the Turken. The name "Turken" arose from the mistaken idea that the bird was a hybrid of a chicken and the domestic turkey.

They make for a good dual-purpose utility chicken. They only have about half the feathers of other chickens, so they are easier to pluck if raised for meat. They also lay a respectable number of eggs. They are very good foragers and are immune to most diseases, plus they are pretty fun to look at!

**Orpington**

Orpington chickens were developed in the town of Orpington, England of all places! During the late 1800s, William Cook wanted to create a new breed that was dual purpose, but had white skin, which the British preferred for meat. Within 10 years, Orpingtons were a favorite in both England and America, and came in a variety of colors – black, white, buff, jubilee, and spangled.

Orpingtons lay about 200 eggs per year. If you're thinking about adding some to your flock, we suggest the Buff Orpington. They are known for being very docile – they make great pets!

**Plymouth Rock**

Developed in America in the middle of the 19th century, this breed of chicken is historically the most popular in the United States. Up until WWII, no other breed was kept as extensively as the Plymouth Rock. The original birds were all of the Barred variety – with black and white stripped plumage – and other color varieties were developed later. The popularity of this duel-purpose breed came from its qualities as an outstanding farm chicken: hardiness, docility, broodiness, and excellent production of brown eggs. This chicken is usually what comes to mind when you think of the chickens Grandma used to keep!

**Rhode Island Red**

Rhode Island Reds are a great choice for beginner chicken-keepers, or expert small flock keepers alike! Developed in Massachusetts and Rhode Island in the late 1800s, these birds are a hardy, dual purpose breed. They are very low maintenance, and can tolerate less than favorable conditions. Hens lay about 5 – 7 eggs per week.

**Speckledy aka Speckled Ranger**

Are you ready for your new favorite breed? The Speckledy is a modern hybrid, resulting from a cross between a Rhode Island Red rooster and a Marans hen.

They are elegant in build, with feathering that resembles a Cuckoo Marans. The feathers are silky, soft to the touch, and quite abundant and fluffy. They have pale bay eyes, pale legs, a medium-sized single comb, and small earlobes and wattles. They are a docile, easy to tame, and easy to handle bird. They are good foragers and well suited to a free-range environment.

Speckledys are very good layers. Hens will lay 250-270 chestnut brown eggs per year, which are often speckled. Their eggshells are particularly strong and the yolks are a deep yellow. They may produce less than some other hybrids, but they will keep your egg cartons full!

**Sussex**

This breed has ancient connections going all the way back to 43 A.D., when the Romans invaded Britain. They grew a reputation of being the finest poultry in Britain, and reached America in 1912. They are a dual-purpose breed and will put on fat very easily, so be careful in feeding them too many treats! If they become too overweight, you will see a decline in egg production. Sussex chickens are a wonderful breed for a small farm or homestead, being active and all-around an excellent breed for meat and eggs. Hens lay an average of 250 light brown eggs each year, and come in three recognized color varieties: Speckled, Red, and Light.

**Welsummer**

Welsummers are a Dutch breed of domestic chicken, developed in the 1920s. It is a light, friendly, and intelligent breed, with rustic-red and orange color. Hens lay large eggs, that are a dark, terracotta brown, and often speckled. Roosters are considered to have the "classic rooster" look, and often used in media.

**Wyandotte**

Developed in the 1880s, Wyandottes are named after a Native American tribe prevalent in parts of upstate New York and Ontario, Canada. They are thought to be developed from the Dark Brahma and Spangled Hamburgs.

The Silver Laced Wyandotte was the original color recognized, but since then black, blue, buff, buff Columbian, Columbian, golden laced, partridge, and silver pencilled have been added as recognized color varieties.

Wyandottes are friendly, calm, and cold hardy. Hens lay on average 200 light brown eggs per year. They make excellent setters and mothers.

**Ancona**

Ancona chickens originated in Italy and are named after the capitol of the Marche region. Anconas were developed in to their present form in England in the 19th century. They were bred to have very consistent plumage. About 1 out of every 3 beetle black feathers has a V-shaped white tip on the end.

Hens are good layers of white eggs and lay about 220 per year. This breed is typical in personality of Mediterranean breeds: rustic, lively, hardy, and ranging.

**Andalusian**

Andalusian chickens are indigenous to Spain. Also called Blue Andalusians, they commonly have slate-blue colored plumage, but depending upon genetics, they can also be off-white or even black. Like other Mediterranean breeds, they have white earlobes. Their light body shape and their large pointed combs make them well-suited for warmer climates. Andalusians are very active foragers, so think twice if you keep your poultry in a coop and run. They do not do well in confinement, and thrive in a free-range environment. Hens lay about 165 white eggs per year.

**Cinnamon Queen**

The Cinnamon Queen is a modern day production breed that lays brown eggs. They are a cross between a Rhode Island Red rooster and a Rhode Island White hen. At hatching, cockerels are a different color than the pullets so you can be sure of what you are getting—no surprise roosters! They are also known as Golden Comets.

Cinnamon Queens were developed specifically for their prolific egg laying ability. Pullets will lay 250-300 eggs per year and start much sooner than heritage breeds. These girls are a perfect fit if you're looking to start a small egg farm or just want a ton of eggs!

**Holland**

In 1934, white eggs brought premium prices at market because it was believed that they had a better, more delicate flavor. Most of America's eggs were produced by small farms at the time. Small farmers prefer dual-purpose chickens because they provide a source of meat as well as eggs. Because dual-purpose chicken breeds tend to lay brown eggs and white egg-laying breeds available at the time were light-weight and not well fleshed, this prompted Rutgers Breeding Farms to set about producing a dual-purpose breed that would lay white eggs. Thus, the Holland was created.

Hollands are great backyard chickens! They are calm, good foragers, and the hens will raise their own offspring. They are also great table birds and lay plenty of large eggs. You can also enjoy that they are helping to conserve what is likely the rarest living breed of American chicken!

**Leghorn**

Is it pronounced "Leghorn" or "Leghern" ? Either way, these birds are great! This breed was developed simultaneously in England and the U.S. in the 1850s, with ancestry tracing back to birds in Northern Italy. Leghorns are very active birds – scratching and foraging the day away. They are hardy and easy breeders, but are mostly known for their egg production. You could easily get 280 eggs in a year, even up to 300! Many of the white eggs you see in grocery stores are produced by this breed of chicken.

**Minorca**

Minorcas are a Mediterranean breed of domestic chicken, and are in fact the largest fowl from this region. They have a greenish-black glossy plumage, and very large, bright red combs and wattles. These help with dissipating heat. They also have very large, almond shaped, white earlobes, common to other Mediterranean fowl.

Minorcas are not broody, but excellent layers of large, white eggs. They are very hardy and rugged, taking well to free range conditions.

**Dutch Bantam**

Originating in the Netherlands, the Dutch Bantam is a true bantam breed. They are one of the smallest bantams, only weighing in at about 15 ounces. They can fly rather well because they are small but have large wings.

Dutch Bantams are especially hardy and good layers for their size. Hens lay about 160 cream colored eggs per year, although they are small. They have friendly temperaments and make great family pets!

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**Pekin Bantam**

Pekin Bantams are of Chinese origin and are alleged to have been looted by British soldiers from the private collection of the Emperor of China. They are round birds and have so many feathers that you practically can't see their feet and legs!

Roosters weigh around 23 ounces and hens weigh around 20 ounces. There is some debate over whether Pekins are a true bantam breed or are just miniature Cochins. Pekins come in a variety of colors. They are docile birds, and with some handing, can make great pets! While hens don't produce many eggs, they are broody and make good mothers (egg adoption anyone?).

**Rosecomb Bantam**

The Rosecomb Bantam is a true bantam breed, meaning it does not have a larger counterpart. It is one of the oldest bantam breeds, developed in England during the 14th century. Roosters weigh about 26 ounces and hens weigh about 22 ounces.

They are kept mainly for exhibition and are generally bred for their appearance. Officially recognized colors include black, white, and blue. Unofficially, there are many more feather varieties. They have a very large comb and earlobes for their body size. They stand tall, alert, and proud, and have an "aristocratic" way about them. Hens lay one tiny cream-colored egg each week.

They are docile in nature, but not at all friendly. In fact, these birds are high maintenance, and only recommended for serious poultry connoisseurs. They are not the best fit for casual poultry

**Scots Dumpy**

Scots Dumpies are an ancient Scottish breed of chicken. Evidence suggests they existed 700 years ago. The “Dumpy” in the name refers to a dwarfing gene that causes them to have very short legs and waddle as they walk.

Because of their shorter legs, the Scots Dumpy can’t scratch up your landscaping! They are also excellent egg producers, broody, and good mothers. They are docile and the roosters make very timely alarm clocks! While all roosters will crow, the Scots Dumpy rooster is more inclined to crow at their first sign of dawn.

**Barbezieux**

The Barbezieux originated in France during the Middle Ages. These birds are impressive in the coop and on the table!

Adult birds weigh 9 to 12 pounds, and roosters can grow up to 2 feet tall. They are considered the tallest chicken in Europe. They have iridescent, beetle black feathers, and an oversized comb and wattles. They have white skin and blue legs. Hens lay a good amount of white eggs, should you want to keep some in your flock as layers. Their meat is ultra-premium. Some foodies claim it's even better than the famous Bresse! Barbezieux meat is firm and has a very distinct flavor, with notes of wild game. The roasted skin is golden and has a light wheat scent.

**Basque Chicken**

The Basque region of Spain and France has a rich history and culture that has endured thousands of years in the harsh and rugged terrain. The Basque people view themselves as independent and apart from the countries where they reside. Just like the people of the region, native chickens thrive under conditions that their less hardy cousins would struggle with.

The legs, feet, and skin of these birds are yellow. They have a bright red single comb and narrow, pointed red earlobes. They are found in five color varieties: Beltza (black), Gorria (red), Lepasoila (naked-necked, red-brown), Marraduna (golden cuckoo) and Zilarra (black-tailed white).

This breed has recently become a favorite of homesteaders in the United States and Canada. They are ideal for free ranging, given their hardy ancestry and excellent foraging skills.

Basque Chickens are medium-sized, with roosters growing as large as 9 pounds, giving a good amount of meat for the table. Hens lay about 200 – 220 large brown eggs per year. Their best quality, however, is their winning personality. They are very friendly and enjoy human company, allowing themselves to be picked up without a fuss.

**Bresse**

The Bresse is hailed as the best tasting chicken in the world. Similar to French Champagne, birds must be raised within the legally defined area of the historic region of Bresse, in eastern France. To maintain the strictest quality standards, the raising and selling of Bresse chickens is rigidly controlled by the French government. There are rules about how much land they must have access to, what they must be fed, and how they must be processed. There are only about 200 breeders that producing 1.2 million birds annually.

When you purchase a Bresse chicken it will have a leg band to prove authenticity. It will also cost you over $20 per pound!

Bresse chickens are colored like the French flag – red comb, white body, and blue legs.

**Cornish**

The Cornish was an epic fail in terms of serving its original purpose. Sir Walter Raleigh Gilbert of England developed the breed, originally naming it the "Indian Game." He intended to combine the power of an Aseel gamebird with the speed of an English gamebird. What he got was a bird with neither of those qualities.

Cornishes were (misleadingly) marketing in the 1800s as an excellent all around bird, despite being "nearly the worst domestic fowls for ordinary use." In the early 1900s, breeders renamed it the "Cornish" and found two niche markets. Due to the muscular nature of the breed, young birds could be harvested early to produce a tender and flavorful one-pound bird: the now well-know "Cornish Game Hens." They are also ideal to cross with American breeds to produce extremely fast growing market poultry.

**Gallina di Saluzzo**

The Gallina di Saluzzo, Italian for "the white hen of Saluzzo," is from the Piedmont in Italy. It is a rustic breed and was once widespread throughout the region. Small family farms are common in the area, and the animals traditionally raised there were intended for family consumption. With the increase of intensive and commercial agriculture, this breed almost disappeared completely, along with other pasture-raised breeds in the area. Recovery efforts began in 1999 to revive the breed because of it's historical significance to the region.

The Gallina di Saluzzo is a dual-purpose breed. While hens lay about 180 white eggs per year, this breed really shines on the dinner table. Due to extensive breeding and very specific feeding techniques, the quality of their meat is exceptional.

They are a classic looking chicken: all white feathers with a red comb and wattles, a yellow beak, and yellow feet. They weigh about 4 – 6 pounds, making them a medium to small-sized bird.

**Gournay**

"Le Poule de Gournay," or the Gournay chicken, is from the upper Normandy region of France. It has ancient ancestry that may date back to the age of Vikings. These birds weigh 4 – 7 pounds and have a round body and small head. Their feathers are evenly mottled black and white. They have orange eyes and a thick beak, and a well-developed breast with delicate and flavorful meat.

Hens are sweet but will go broody. They lay around 3 white, extra-large eggs per week. The Gournay is easy to tame and handles confinement well, making them a rare but excellent choice for backyard chicken keeping.

The Gournay, like many traditional European breeds, suffered during World War I and II. They nearly went extinct, but with the help of local enthusiasts in the early 2000s, there are now around 15,000 of these birds in France.

**Ixworth**

The Ixworth was created in the 1930s England, by Reginald Appleyard. Appleyard is better known for developing the Silver Appleyard Duck. In the Ixworth, he envisioned the ideal dual-purpose breed that would be an active forager, produce eggs, and make a hearty meal.

Both hens and roosters have a stocky body, pure white feathers, white skin, and pea comb. While they are better designed for the table, hens also lay a decent number of tinted eggs.

**New Hampshire**

New Hampshire chickens have only been around since the 1930s and are closely related to the classic Rhode Island Red (RIR). Starting with RIRs, breeders were very selective and intensified traits of early maturity, rapid full feathering, and production of large brown eggs. These birds are a rich chestnut color, slightly lighter in shade than RIRs.

New Hampshires are a dual-purpose breed, but are intended more for the table than for egg production. They prefer to free range and are known to be competitive with the rest of the flock.

**Norfolk Grey**

The Norfolk Grey is a utility breed developed in England in 1910, by Frederick Myhill. Originally, the breed was called “Black Maria”.

During World War I, Myhill had to leave his flock to free range while he left for military service. When he returned home, he discovered that his birds had cross bred with other breeds, and he had to start over again. While he successfully did so, Black Marias did not gain in popularity, so Myhill had the name changed to Norfolk Grey. This breed all but died out in the 1970s. A private flock of only 4 birds was able to revive it!

Norfolk Greys are a heavy breed, but not as large as other birds in that category. They have a single comb, black eyes, slate legs, with black body feathers and silver striped hackles.

This breed does well free ranging. Roosters weigh 7 – 8 pounds and hens weigh 5 – 6 pounds. They produce a good roast for the table, and hens lay 150-220 pale brown eggs per year.

**Red Cap**

The RedCap is an egg-laying breed from England named after their very large rose comb. It is one of the older English breeds, but exact time is unclear. The RedCap was bred so much for utility that they are lacking in appeal. Their coloring, wild temperament, and generally unrefined quality led them to fall out of favor by 1900. The breed has red plumage tipped with a blue-black, half-moon shaped spangle and leaden blue colored legs.

During the early to mid-1800's the Redcap chicken was considered one of the most profitable fowls a farmer could have. They have delicate meat, and even though they have red earlobes, can lay 150-200 white eggs.

**Red Shaver**

The Red Shaver is a sex-linked breed from Canada. Female chicks are a reddish-brown color with white underfeathers, while male chicks are white with a few red markings on the feathers. They are a dual purpose breed with a reputation for being quiet and calm. Hens lay up to 300 large brown eggs per year! Because they are Canadian, they are very well adapted to cold.

**Vorwerk**

The Vorwerk was developed in 1900 by Oskar Vorwerk in Hamburg, Germany. His goal was to create a medium-sized, dual-purpose breed with the same feather pattern as the Lakenvelder.

Vorwerks are hardy, adaptable, alert, and active. They mature quickly, are pretty good at flying for a chicken! This makes them great candidates for free range flocks.

Birds are typically 4 – 8 pounds and hens lay about 170 large cream eggs per year. They are a golden buff color, with solid black head, neck, and tail. They are often confused with Golden Lakenvelders, but are a separate breed. This is perhaps the reason why Vorwerks never gained popularity, therefore they are rare outside of Europe.

**Old English Game**

The Old English Game breed is one of the oldest breeds of fowl, having been introduced to England by Romans in the 1st century! Although their origins were in the fighting ring, today they are raised for their exotic appearance and only for show. They have compact, muscular bodies with feathers that are hard, glossy, and sit tight along the body. They are known to have fearless eyes and an indomitable spirit.

Old English Games come in a variety of colors. Both hens and roosters have large and distinctive tail feathers. Hens make excellent brooders, although they can be overly aggressive as mothers. They can tolerate extreme climates, are good foragers, and do well in free range situations. But watch out! Their excellent stamina and flying capabilities make them crafty escape artists. Many Old English Games live as long as 15 years or more.

**Brussbar**

The Brussbar is a dual purpose breed, developed by Professor Punnett and Mr. Pease at Cambridge in the first half of the 20th century. They were looking to create an autosexing breed with the characteristics of a Light Sussex, the most popular breed at the time. Brown Sussex and Barred Rocks were used initially to create the autosexing plumage with utility strains of Light Sussex added to improve productivity. The breed was officially standardized in 1952.

Birds are large and well-built, similar to a Sussex. They have a distinctive crele plumage, a copper and gold body color with barring patterns throughout. Originally, the Brussbar came in both a gold and silver variety, though now only the gold variety can be found. Brussbars never caught on in popularity and the breed barely survived the 1960s. Only one farm was keeping this breed alive! They remain very rare today.

**California Grey**

Developed in California in the 1930s by James Dryden, professor of poultry science. He wanted a dual-purpose hen laying large white eggs that remained in her egg-laying prime for longer than 2 years. He crossed a Barred Plymouth Rock rooster with a White Leghorn hen, which resulted in a naturally autosexing breed with grey barred plumage.

Birds are between 4 – 6 pounds, which makes them too large to appeal to commercial egg producers. They were also never recognized by the American Poultry Association. This means they never enjoyed popularity and today are a rare find.

If you manage to have this breed in your flock, you're a lucky one! These birds are calm, good winter layers, and given the intent of their creator, should produce lots of eggs in their lifetime!

**Catalana**

Catalanas were developed near Barcelona in the district of Catalonia, Spain. It was introduced to the rest of the world at the 1902 World's Fair held in Madrid. They are a hardy, dual-purpose breed, with the style, alertness, and forage abilities typical of Mediterranean breeds. They lay large white eggs and rarely are broody. Catalanas are noted for being very heat tolerant.

**Dorking**

The history of the Dorking is similar to that of the Sussex. An ancient breed with ties to the Roman Empire, the Dorking was developed to be a superior table bird. As a backyard poultry keeper, this breed would make an excellent dual-purpose bird! Hens make excellent winter layers, and are exceptional mothers. They welcome chicks from other hens and tend to look after chicks far longer than other hens.

**Langshan**

Langshans originated in China near the Yangtszekiang River and made its way to England in 1872. The breed is valued for being a large bird, with quality meat, that lays a high volume of dark brown, purplish eggs.

Langshans are hearty birds and good foragers. They have tight feathering and can fly better than most other chickens. Hens are not dependable sitters but make excellent mothers once the chicks have hatched.

**Marsh Daisy**

The Marsh Daisy is a very rare breed originating in Lancashire, England, and has not made a name for itself in other countries. It's a bird with a fancy name and a practical nature. These birds are slow to mature, but once grown, are very hardy and excellent foragers. They flourish in free range environments. Hens lay about 200 cream colored eggs each year.

The Marsh Daisy chicken may be one of the rarest chicken breeds worldwide. It never achieved popularity abroad, was never recognized by the American Poultry Association and is little known or seen outside the UK.

**Norwegian Jaehorn**

The Jaehorn is the only breed of domestic chicken indigenous to Norway. They were developed in 1920 near the town of Stavenger. They have only recently made their way to North America and are still a rare find in the U.S.

They come in two colors: dark brown and light brown. Hens can lay an impressive 215 white eggs per year. Jaehorns are small, hardy, and active birds. They are great flyers due to their size. So be aware, your chickens may fly the coop!

!

**Penedesenca**

The Penedesenca originated in the region of Catalonia, in Spain. They are named after the town Vilafranca del Penedes and were developed from native backyard birds.

This breed is known for the very dark brown eggs that the hens produce. They are said to be the darkest brown of any breed. They come is a few color varieties: Black, Crele, Partridge, and Wheaten. They have white earlobes, red wattles, and an unusual carnation comb.

This breed is extremely rare. In fact, they almost went extinct! In the 1980s, some breeders dedicated themselves to reviving the Penedesenca.

**Rhodebar**

The Rhodebar is a breed that we wish wasn't quite so rare! Hailed as "an absolute gift for the small poultry enthusiast", these birds have so many great qualities.

Originally created at the University of British Columbia in the 1940s, during the autosexing breed development craze, the Rhodebar involves a cross of Rhode Island Reds and Plymouth Barred Rocks. A version of the breed was also created in the U.K. by crossing a Danish strain of Rhode Island Red with Golden Brussbars.

Rhodebars are autosexing, meaning males and females have different coloring at the time of hatch. Males are yellow, and females have dark stripes of barring down their backs. This makes them easy to differentiate, so you stand an excellent chance of knowing if you've got future hens or roosters in the bunch.

This is a dual-purpose breed. Hens are great layers of brown tinted eggs. You can expect about 180-200 eggs per year. Birds weigh between 6 – 9 pounds and dress out nicely for the table.

Because the Rhodebar was developed just before the dawn of commercial hybrid breeds, it only enjoyed brief popularity, and is now very rare. Those who continue to breed and raise Rhodebars are very devoted to maintaining the exceptional qualities of this breed.

very defensive mothers.

**Twentse**

Twentses (Dutch), also known as Kraienkoppes (German), are a large breed of chicken from an area spanning between Germany and the Netherlands. They are rumored to be the result of Leghorn and Malay crosses and are sporty, ornamental birds that also have good egg production. Hens lay about 200 off-white eggs per year.

They have small wattles, earlobes, and walnut comb—all bright red in color. This rare breed is an excellent forager in both free range and confined conditions.

bones, even black organs! They have a gene that causes hyperpigmentation (Fibromelanosis). While their blood is still red, it is very dark. Due to their exotic appearance, in Asia their meat is considered to have mystic powers.

They are friendly birds, hardy, low maintenance, and easy to handle. Hens lay a moderate amount of cream colored eggs that are relatively large in proportion to their body size.

Consider yourself lucky if you can find a bird of this breed! They are extremely rare and can be expensive.

**Cubalaya**

The Cubalaya is the only breed developed in Cuba. It is descended from Sumatra and Malay birds brought to Cuba from the Philippines. They were selectively bred to be impressive in appearance, with a courageous expression. Roosters have flowing hackle feathers and a “lobster tail” – a downward angled tail with lavish feathering. Their look is truly unique to the breed.

These birds are very tame, with a friendly and curious disposition. They very heat tolerant, for those of you with long hot summers! The hens are reliable layers, lay small eggs and are good brooders.

## SELF CHECK 1.1-1

### Chicken breeds identification

1. _________Which breed of chicken developed in California in the 1930s by James Dryden, a dual purpose hen crossed by a Barred Plymouth Rock rooster with a White Leghorn hen?
   a. California Grey c. Plymouth rock
   b. White leghorn d. Cubalaya
2. _________Which breeds of chicken that are a large from an area spanning between Germany and the Netherlands, have good egg production and an excellent forager in both free range and confined conditions?
   a. Twetse c. Leghorn
   b. Rhodebar d. Langshan
3. These are a dual-purpose breed, but are intended more for the table than for egg production. They prefer to free range and are known to be competitive with the rest of the flock.
   a. New Hampshires c. Red cap
   b. Penedesenca d. Cubalaya
4. What is the smallest bantams only weighing in at about 15 ounces and is considered as true Bantam breed?
   a. Dutch Bantam c. Red Comb Bantam
   b. Pekin Bantam d. Bantam
5. Which breed that are a great choice for beginner and are a hardy, dual purpose breed. They are very low maintenance, and can tolerate less than favorable conditions. Hens lay about 5 – 7 eggs per week.
   a. Rhode Island Red c. Langshan
   b. Leghorn d. Dorking

## ANSWER KEY 1.1-1
## Chicken breeds identification

1. a
2. a
3. a
4. a
5. a

**JOB SHEET 1.1-1**

**Chicken breeds identification**

| |
|---|
| **Title: Identifying breeds of Chicken** |
| **Performance Objective:** At the end of this module the student should be able to identify breeds of chicken |
| **Supplies/Materials:** PPE, broiler, layer, dual purpose |
| **Steps/Procedures:**<br><br>Wear the PPE<br>Observe the physical appearance of chicken<br>Look for their differences<br>Identify, which is broiler, layer and dual purpose<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 1.1-1

## Chicken breeds identification

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee .. | | |
| 1. Wear the PPE? | | |
| 2. Observe the physical appearance of chicken? | | |
| 3. Look for their differences? | | |
| 4. Identify, which is broiler, layer and dual purpose? | | |
| 5 Do housekeeping? | | |

**INFORMATION SHEET 1.1-2**

**Healthy Chicks Selection**

**Learning Objective:** After reading this information sheet, you should be able to select healthy chicks.

**Introduction:**

**How to Pick a Healthy Chick**

Chick Days are in full swing and if you walk into any farm and ranch store, chances are good that you'll be greeted with fuzzy, adorable peeping chicks. Who can resist?! But before you scoop up your chicks, you'll want to make sure you're choosing the healthiest of the flock.

Here are ways to spot a healthy chick:

- They're alert and active – may be cheeping softly, looking for food, and will move away from you when approached. Chicks that aren't well will appear lethargic and constantly sleepy, and may not try to move away when approached
- If chicks are happy, healthy and warm, they won't huddle together when awake
- Bright-eyed – chicks with a blank stare, crusted eyes or always sleepy may not be healthy
- Look for a beak that is not crossed over or broken – birds with beak issues will have problems eating and drinking

- Healthy feathers – unless you're buying an older chicken during molt, chickens and chicks shouldn't be missing feathers
- Straight legs, feet and toes – an unhealthy chick may have difficulty walking or have poor posture with its neck retracted into its body
- If a bird is acting dull, withdrawn or hunched over, it could indicate a serious problem

You don't always have to buy your birds from a store, sometimes you can purchase them directly from a breeder, or a friend, or someone looking to re-home their hen. These tips apply to both chicks and full-grown chickens and the key is to make sure you purchase your birds from a reputable source to avoid serious health problems with your flock. If you're going to buy from a breeder, here is a list of the best breeds for beginners.

## SELF CHECK 1.1-2

## Healthy Chicks selection

Enumeration:

1. What are the different ways of spotting healthy chicks? Give at least 5.

**ANSWER KEY 1.1-2**
**Healthy Chicks selection**

1. Chicks are alert and active
2. Bright-eyed chicks
3. beak that is not crossed over or broken
4. Healthy feathers
5. Straight legs, feet and toes

**JOB SHEET 1.1-2**

**Healthy Chicks Selection**

| |
|---|
| **Title: Selecting Healthy Chicks** |
| **Performance Objective:** At the end of this module the student should be able to select and identify healthy chicks. |
| **Supplies/Materials:** PPE, chicks |
| **Steps/Procedures:**<br><br>Wear the PPE<br>Examine the chicks' physical appearance<br>Describe its eyes, feathers beak and legs<br>Look for its uniformity<br>Identify if it's healthy<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

# PERFORMANCE CHECKLIST 1.1-2

## Healthy Chicks Selection

| **Performance Criteria** | **YES** | **NO** |
|---|---|---|
| Did the trainee .. | | |
| 1. Wear the PPE? | | |
| 2. Examine the chicks' physical appearance? | | |
| 3. Describe its eyes, feathers beak and legs? | | |
| 4 Look for its uniformity? | | |
| 5 Identify if it's healthy? | | |
| 6 Do housekeeping? | | |

## INFORMATION SHEET 1.1-3

### Determining suitable site for chicken house

**Learning Objective:** After reading this information sheet, you should be able to determine suitable site for chicken house.

**Introduction:**

Avoid low-lying areas near streams with flooding potential. Preferably, the topography will allow the long axis of the poultry house to be located in **an east-west direction**. This helps to minimize the amount of direct sunlight that would enter through the sidewalls of the houses.

**Housing**

We distinguish three forms of chicken farming:

- extensive farming
- intensive farming
- semi-intensive farming

When chickens are free to roam and scavenge, we talk about *extensive*, free-range chicken farms. The level of capital and labor investment is low. Housing is not important.

*Intensive* systems, developed for specialized breeds, are estimated to be in use for about 30% of the poultry population. These are mainly found in and around urban areas with good markets for eggs and chicken meat. Intensive chicken farms require more investment of both capital and labor, e.g. special poultry houses with runs or roaming space. Flock sizes in intensive production are normally in the thousands. This has been made possible by research developments in artificial incubation, nutritional requirements and disease control.

In the *semi-intensive* production system, also known as backyard production flocks range from 50 to 200 birds. A lot of techniques and expertise developed in intensive systems can be applied in semi-intensive poultry raising systems, adapted to the adequate scale.

In both the semi-intensive and intensive production systems, housing is very important for optimal production levels.

**Free-range chickens**

- In the free-range system, chickens are free to roam the farm in search of food. Eggs are laid outside in simple nests and are mainly used to maintain chicken numbers. In many cases, up to 75% of the eggs have to be hatched because
- the mortality rate among baby chicks is high. Few eggs remain for consumption and the chickens themselves do not give much meat.
- 
- The advantages of this system are that little labour is needed and waste food can be used efficiently. Very low costs can offset low production levels so that keeping chickens around the house can be profitable if certain improvements are made.
- 
- The free-range system is most suitable if you have a lot of space, preferably covered with grass. At night, the chickens can be kept in any kind of shelter, as long as it is roomy, airy and clean. This will minimize the loss of chickens to illness or theft. If you have enough space for the chickens to roam freely, a mobile chicken house is best.

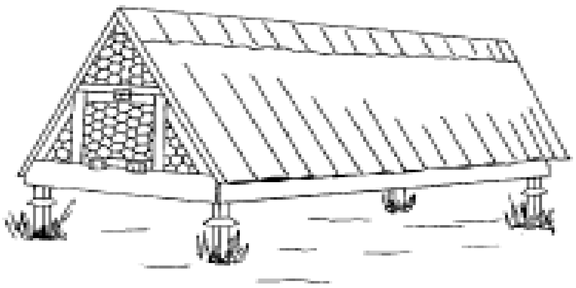

Figure 3.1: A simple mobile chicken house

- The spread of infection by parasites in chicken feces can be prevented by using a raised night shelter with an open floor made of chicken-wire, wooden slats or bamboo sticks 5 cm apart. This will also keep the chickens safe from predators. If you want to maximize the number of eggs, train mature layers to use laying nests in the chicken house early in the laying period. Place the laying nests in the chicken house before the chickens start laying, and keep them in a bit longer in the morning. Remember to provide fresh drinking water.

- To limit mortality among baby chicks in the free-range system, take steps to protect the mother hen and the chicks from predators, thieves and rain. Put them in a simple, separate shelter that is roomy and airy and can be closed securely. Draughts and low temperatures during the first few days are particularly dangerous for the baby chicks. Although a run is handy, it is also risky, due to possible worm infections. It is important to move the run regularly, especially in wet weather.
- 
- Fold units are very suitable mobile housing units for young chicks. These cages can house 20 young hens, and contain feeders, drinkers and a perch. Obviously, you will need enough space to move the fold units around regularly.

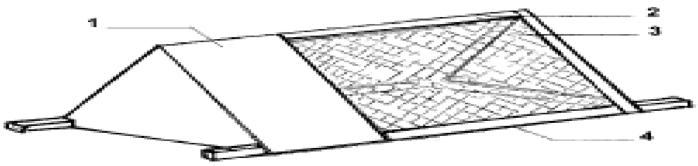


Figure 3.2: Fold unit for housing young chicks. 1. boarded section 2. wooden framework 3. wire mesh 4. wired floor.

In areas where dogs or predators are a problem, it might be worth building a shelter well above ground level (e.g. 1.20 m high). Tin rat baffles around the supporting poles will keep out rats and other small animals. The baffle must fit tightly to keep even the smallest rodent from climbing between the baffle and the pole.

Always ensure a steady supply of clean, fresh drinking water. Give your chicks extra feed, including greens which are rich in vitamins. If possible, vaccinate the chicks against the most common contagious viruses, such as Newcastle Disease.

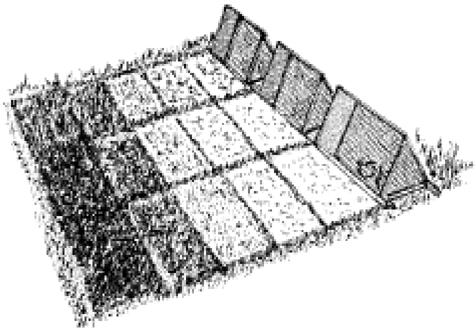

Figure 3.3: Systematic daily movement of fold units

*Advantages of the free-range system*

- Exercise in the open air keeps chickens healthy.
- Feed, even if it is not well balanced, presents few problems.
- Parasitic infections can be kept to a minimum if there is enough space.
- Little or no labor input is needed.
- The chickens help limit the amount of rubbish in a productive way.
- The direct costs of the system are low.

*Disadvantages of the free-range system*

- Free-range chickens are difficult to control.
- The chickens, especially young chicks, are easy prey for predators.
- Chickens may eat sown seed when looking for food.
- A large percentage of the eggs can be lost if the laying hens are not accustomed to laying nests.
- Mortality rates are usually high.

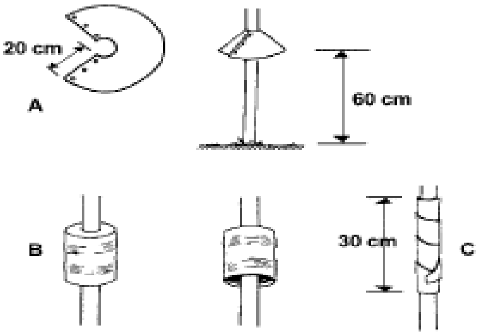


Figure 3.4: Examples of rat baffles. A. metal collar B. metal can upside down C. metal band around post

**Small-scale housing**

In both the intensive and semi-intensive production systems, housing becomes very important for improving working conditions and minimizing risks. Adequate housing facilitates the feeding and egg laying and thus is a primary condition for optimal production levels.

If you decide to keep your chickens in a special poultry house, consider the following:

- You will certainly incur extra costs.
- Make sure that necessary materials are locally available.
- Should your chicken have a run? If you opt for a run, check that there is enough space to change its position regularly.
- Decide whether to continue to breed own chicken stock or to buy new stock. If you breed your own stock, you need to build more houses for separating chicks of different ages.

**Optimizing climate in the house**

Chickens can tolerate high temperatures but react negatively if they are too warm. Try the following as guideline when designing the poultry house.

Build the house in an east-west direction, so the chickens are less exposed to direct sunlight. Place the house where there is grass, herbs or other vegetation. Plant trees around it to keep its roof shaded. Make sure that the roof has a large overhang of 90 cm or more to limit direct sunlight and keep out the rain. Build the roof as high as possible above the floor. The chicken house will then be cooler and better ventilated.

Keep the bottom 50 cm of the side walls closed and the rest open to allow enough fresh air into the house. Close the top part of the sidewalls with chicken wire or some other suitable material. A chicken house can have a corrugated metal roof, but in a sunny place, this will certainly overheat the house. In this case cover the roof with leaves or some other material. A disadvantage of this is that rodents like rats and mice can nestle in the covering. Do not keep too many chickens in the chicken house. Doing so can make the house too warm and help to spread parasitic infections. In hard-floor housing, there should be no more than 3 chickens per square meter. In houses with wire netting or slatted floors, a higher chicken density is possible.

Finally, to stimulate feeding in cooler weather, turn on a light in the house before sunrise and after sunset. This also helps to keep a steady level of egg production.

## SELF CHECK 1.1-3

## Determining suitable site for chicken house

**TRUE OR FALSE**

1. ____________________ Avoid low-lying areas near streams with flooding potential.
2. ____________________ In the free-range system, chickens are free to roam the farm in search of food. Eggs are laid outside in simple nests and are mainly used to maintain chicken numbers
3. __________________ the topography will allow the long axis of the poultry house to be located in an east-west direction. This helps to minimize the amount of direct sunlight that would enter through the sidewalls of the houses.
4. _________________ The spread of infection by parasites in chicken feces can be prevented by using a raised night shelter with an open floor made of chicken-wire, wooden slats or bamboo sticks 5 cm apart.
5. _________________ In both the intensive and semi-intensive production systems, housing becomes very important for improving working conditions and minimizing risks.

## ANSWER KEY 1.1-3

## Determining suitable site for chicken house

1. True
2. True
3. True
4. True
5. True

## INFORMATION SHEET 1.1-4

## Chicken house design preparation

**Learning Objective:** After reading this information sheet, you should be able to prepare design for chicken house.

**Introduction:**

One of the main factors of whether your chicken farming venture in the Philippines will succeed or not is going to be the quality of your chicken coop or chicken house.

It is so because a poorly design chicken coop can expose your poultry flock to weather extremes, lead to overstocking which leads to antisocial behavior and greater stress of your birds, and expose your flock to predators.

### Make Sure the Chicken Coop Is Predator-Proof

The very first thing to keep in mind is to keep your chicken house completely predator-proof. This means that not only the sides and the roof should be completely protected, but also the ground since some predators will try to burrow their way to reach your flock.

If you have a chicken run that has been fenced with barbed wire, ensure that there are no holes beneath the wire mesh fencing. Also, as standard fencing wires generally have large holes that can allow various predators to get it, it is advisable to reinforce them with a chicken wire with smaller holes that will not allow any dangerous animal through them.

Additionally, you can also position the chicken coop or chicken house among tall trees so as to protect the birds from flying predators especially if they are still young.

### Protect Your Chicken House from Rodents

While predators are a threat to the chicken directly, rats and rodents burrow their way into chicken houses to access chicken feed and other leftover food. On top of that, they also spread diseases, and as such, it is best to keep them as far from your flock as possible.

Some of the things you can do to make your chicken coop rat-proof, include cementing the flooring or installing a small mesh fence on the ground below the coop.

**Chicken Houses in the Philippines Need to be Breezy**

The Philippines is generally a hot country, and chickens are birds with plumage designed to protect them from excessive cold. Because of this, your chicken house must be well ventilated and a little breezy.

Have low air pressure coming into your chicken house in the Philippines. However, don't make it too draughty. Draught is simply too much cold air coming into the chicken coop which makes it uncomfortable for the birds during the cold months.

And, as long as the conditions are not too draughty, the chickens will be able to withstand the cold.

**Design the House for Easy Cleaning**

It is easy to get carried away and build a fancy chicken house with many nook-and-crannies. However, when designing your coop, keep in mind the fact that you will have to clean it regularly to avoid pestering of bacteria and bugs.

As such, to the maximum extent possible, keep the design simple and easily cleanable.

Separately, also pick equipment that is easy to clean. For example, where possible, have removable droppings trays as they are easier to clean than having to clean the house itself. One of the places you can place them is under the perching poles so as to capture most of the droppings.

**Provide Your Chickens with Roosting Poles**

No matter what farming system you use, but especially in free range chicken farming and native Philippines chicken farming, it is important to set up roosting poles in your chicken house to allow for proper rest of your birds.

The roosting poles or perching poles should be about 2 inches wide with rounded edges. It is advisable to allow for about 10 inches between each of the poles, and about 5 to 10 inches of sideway-space per bird.

Also, the poles should be positioned in the corners of the chicken coop to allow for easy movement of the birds within the chicken house.

If you are space constrained, you can arrange the perching poles in a ladder-like set-up.

**Provide Nest Boxes in the House**

In order to encourage egg laying by your hens, it is advisable to provide one nest box for every 4 or 5 chickens. For maximum efficiency, the nest boxes should be dark and out of the way in a quiet area.

The reason to keep the boxes in the least trafficked place of your chicken house is that hens generally have the instinct to lay their eggs in the safest and quietest possible place they can find.

**Make Sure the Chicken House Is Roomy Enough**

The last thing I will mention here is that you should make sure to build your chicken house roomy enough.

It should be able to accommodate the feeders and drinkers at a safe distance so as to avoid congestion and allow the birds to feed freely without fighting or contaminating the food.

## SELF CHECK 1.1-4

## Chicken house design preparation

**TRUE OR FALSE: Write True if the statement is correct, write False if it is incorrect.**

1. ______ Poorly design chicken coop can expose your poultry flock to weather extremes, lead to overstocking which leads to antisocial behavior and greater stress of your birds, and expose your flock to predators.
2. ______ Some predators will try to burrow their way to reach your flock.
3. ______ If you have a chicken run that has been fenced with barbed wire, ensure that there are no holes beneath the wire mesh fencing.
4. ______ When designing chicken house keep the design simple and easily cleanable.
5. ______ In order to encourage egg laying by your hens, it is advisable to provide one nest box for every 4 or 5 chickens.

**ANSWER KEY 1.1-4**
**Chicken house design preparation**

1. True
2. True
3. True
4. True
5. True

**Learning Outcome # 2 SET-UP CAGE EQUPMENT**

**Assessment Criteria:**

House equipment are installed in line with housing equipment installation design
**Bedding materials** are secured based on availability in the locality
Bedding is prepared in accordance with housing equipment housing design
**Brooding facility** is set-up in accordance with the housing equipment installation design.

**Contents:** House equipment installation
Prepare and secure bedding materials
Set-up brooding facility

**Tools, Materials and Equipment and Facilities**

Farm
Housing
Bedding materials
Brooding facility
PPE (Boots, surgical masks, disposable gloves, overall

**References:**


https://www.thepoultrysite.com/articles/putting-down-perfect-bedding-for-your-poultry

https://www.agriculturediary.com/poultry-farming-poultry-housing-equipment

https://www.farmanddairy.com/top-stories/how-to-set-up-a-brooder-for-baby-chicks/469356.html

## LEARNING EXPERIENCES/ACTIVITIES

### LO # 2 SET-UP CAGE EQUPMENT

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 1.2-1 on House equipment installation | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.2-1 on House equipment installation | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.2-1 on House equipment installation | |
| Read the attached Information Sheet 1.2-2 Prepare and secure bedding materials | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.2-2 Prepare and secure bedding materials | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.2-2 Prepare and secure bedding materials | |
| Perform Job Sheet 1.2-2 Prepare and secure bedding materials | Observe the demonstration of your trainer and check the procedures while performing the operation. |

| | |
|---|---|
| Evaluate performance using performance checklist 1.2-2 Prepare and secure bedding materials | |
| Read the attached Information Sheet 1.2-3 on Set-up brooding facility | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.2-3 on Set-up brooding facility | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.2-3 on Set-up brooding facility | |
| Perform Job Sheet 1.2-3 on Set-up brooding facility | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance 1.2-3 on Set-up brooding facility | |

***After doing this activity, you are now ready to proceed to the next module on Feed Chicken***

## INFORMATION SHEET 1.2-1

### House equipment installation

**Learning Objective:** After reading this information sheet, you should be able to appreciate the importance of house equipment installation.

**Introduction:**

The need for a **Poultry Housing** a) Protection from other climatic extremes such as direct sun, wind, rain and even against theft and attack from natural enemies of the birds such as, fox, dog, cat, kite, snake, etc. The birds also should be protected against external parasites like ticks, lice, mice, etc.
b) Comfort: The best egg production is secured from birds that are comfortable and happy. To be comfortable, a house must provide adequate accommodation, be reasonably cool in the hot weather, free from drafts and sufficiently warm during the cool weather. Above all, provide adequate supply of fresh air and sunshine; and remain dry always. c) Provision of dry condition which are hygienic and do not predispose the birds to diseases and parasites. d) Allowing, as far as possible, for inherent behavior patterns of the birds, and minimizing the effect of social dominance. e) Convenience: The house should be located at a convenient place, and the equipment so arranged as to allow cleaning and other necessary operation as required. f) Provision of accessible food and clean water and for effective disposal of waste. g) Providing condition so that good stockmanship can be practiced.

**Location of Poultry Housing**

In planning a **poultry house**, the location should be taken into consideration. In selecting site for **poultry houses**, the following factors should be considered:

1. **Relation to other Buildings**:- The poultry house should not be close to the home as too create unsanitary condition. On the other hand, it should not be too far away either because this will require more time in going to and fro in caring for the birds. In general, at least three trips should be made daily to the poultry house in feeding, watering and gathering the eggs.

2. **Exposure**:- The poultry house should face south or east in most localities. A southern exposure permits more sunlight in the house than any of the other possible exposures. An eastern exposure is almost as good as a southern one. Birds prefer morning sunlight to that of the afternoon. The birds are more active in the morning and will spend more time in the sunlight.
3. **Soil and Drainage**:- If possible, the poultry house should be placed on a sloping hillside rather than a hilltop or in the bottom of a valley. A sloping hillside provides good drainage and affords some protection.

**Poultry Farming: Poultry Housing and Equipment** The type of soil is also very important if the birds are to be given a range. A fertile well drained soil is desired. This will be a sandy loam rather than a heavy clay soil. A fertile soil will grow good vegetation which is one of the main reason for providing range. If poultry house is located on flat poorly drained soil, the yards should be tiled, otherwise, the birds should be kept in total confinement.

1. **Shade and Protection**:- Shade and protection of the **poultry house** are just as desirable as for the house. Trees serve as a Windbreak in the rainy season and for shade in the dry season. They should be tall, and not very close to the soil. Dwarf tree can become contaminated, makes the soil damp and prevent sunlight from reaching the soil to destroy the germs. One thing a farmer should note is that plenty of sunshine should be available at the site.

**Housing Requirements:**

(1) **Floor Space**: The smaller the house, the more square feet are required for each hen. Bigger pens have more actual usable floor space per bird than smaller pens. The recommendations suggested below might be useful regarding floor, feeders and watering space. For economic production of laying hens, it is always better to keep them in small unit of 15-25 birds. This number can go up to a maximum limit of 250 or so are advisable. When there is a long house, partitioning at every unit should be made to eliminate drafts etc.

**Table 1: Floor space requirement per bird**

| **No.** | **Age (weeks)** | **Floor space per birds(Cm2)** |
|---|---|---|
| 1 | 0-8 | 700 |
| 2 | 9-12 | 950 |
| 3 | 13-20 | 1,900 |

| | | |
|---|---|---|
| 4 | 21-above | 2,300-2,800 |

**Table 2: Feeder space requirement per bird**

| **No.** | **Age (weeks)** | **Feeder space per birds(linear Cm)** |
|---|---|---|
| 1 | 0-2 | 2.5 |
| 2 | 3-6 | 4.0 |
| 3 | 7-12 | 7.5 |
| 4 | 13 and above | 10.0 |

**Table 3: Amount of water required and watering space for chicken**

| **Age (weeks)** | **Water space per chick(linear cm)** | **Amount of water per 1000 birds (liters)** |
|---|---|---|
| 0-4 | 0-6cm | 2.8-4 |
| 5-8 | 1.2cm | 12-14 |
| 9-12 | 10cm | 20-25 |
| 13-16 | 12.5cm | 35-40 |
| 16 and above | 15cm | 45-48 |

(2) **Ventilation**: Ventilation in the poultry house is necessary to provide the birds with fresh air and to carry off moisture. Since the fowl is a small animal with a rapid metabolism, its air requirements per unit of the body is high in comparison with that of other animals. A hen weighing 2kg and on full feed, produces about 52 litres of $CO_2$ every 24 hours. Since $CO_2$ content of expired air is about 3.5%, total air breathed amounts to 0.5 litre per kg live weight per minute. A house that is well tall enough for the attendant to move around comfortably will supply far more air space than will be required by the birds that can be accommodated in the given floor space.

**Poultry Farming: Poultry Housing and Equipment**

(3) **Temperature:** Hens needs a moderate temperature of 50-70oF. Birds need a warmer temperature at night, then they are inactive, than during the day. The use of insulation with straw pack or other materials not only keeps the house warmer during the rainy seasons, but also cooler during the dry seasons. Cross ventilation also aids in keeping the house comfortable during hot weather.

(4) **Dryness**: Absolute dry conditions inside a poultry house is always an ideal condition. Dampness causes discomfort to the birds and also gives rise to diseases like colds, pneumonia. Dampness in poultry house is caused by:- (a) Moisture

rising through the floor, (b) leaky roofs or wall, (c) Rain or snow entering through the windows, (d) leaky water containers, (e) Exhalation of birds.

(5) **Light**: Daylight in the house is desirable for the comfort of the birds. They seem more contented on bright sunny days than in dark, cloudy weather. Sunlight in the poultry house is desirable not only because of the destruction of disease and germs, it also for supplying vitamin D; but also because, it brightens the house and makes the birds happy. Birds do fairly well when kept under artificial light.

(6) **Sanitation**: The worst enemies of the birds, i.e. lice, ticks, fleas, and mites are abundant in poultry houses. They do not only transmit diseases, but also retard growth and laying capacity. The design of the house should be such which admits easy cleaning and spraying. There should be minimum cracks and crevices. Angle irons for the frames and cement asbestos or metal sheets for the roof and walls are ideal construction materials, as they permit effective disinfection of the house. When the wood is to be used, every piece should be treated with coaltar, creosote or any other similar insecticides before being fitted. Used engine oil mixed with wood treatment chemicals can also serve as a good alternative.

**Types of Roofs for a Poultry House**

There are several styles of poultry house with reference to types of roofs:

1) **SHED Types**:- This is the simplest type of poultry house and by far the most useful and practical type of house that can be used under different climatic conditions and for different systems of poultry keeping. The slope of the roof needs only be slight in the plains, while in heavy rainfall, it ought to be sufficiently steep. The shed-roof types of houses may be either portable or stationary. The portable house in generally a small one, not exceeding 8×6 inches while the stationary types can be made of any dimensions.

2) **GABLE ROOF TYPE**:- This type requires more material and labour for construction. Some poultry farmers put a ceiling floor in gable roof houses and use the space in the gable for storage. The type is more suitable in rainfall areas. Here, again, gable type may be stationary or portable.

3) **Combination**:- Such houses have double pitch roofs in which the ridge between the two slopes is not mid-way from front to back. Most of the gable type, the combination roof requires more materials and labor than the shed roof.

**House Construction:**

1. **Roofs**: in most African countries, cement- asbestos sheeting although, very satisfactory and durable is expensive, yet, it is still recommended if the farmer has the capital. Conjugated iron and zinc sheets are equally satisfactory, but the cost is lower than cement asbestos. You can ceil the house with zinc sheets.

2. **Door**:- The door of the poultry house must be on the south, and made of an angle iron frame covered with ½ " mesh Wire netting. The size of the room should be always large enough to allow a man to conveniently pass through.
3. **Windows:**– About one meter block work is recommended as the normal height and the remaining upper part of the wall would be walls to the pillar post. Remember to make the roof overhang at least 18-36 inches out from the wall to cut down radiation through the window opening.

**Poultry Farming: Poultry Housing and Equipment**
**Poultry House Equipment**
The poultry house should be equipped with roasts, nests, feed hoppers, water containers and any other items which is essential for satisfactory production.

- It should be simple in construction
- Cheap
- Movable
- Easily Cleaned
- Easily disinfected whenever necessary

1. **Perches or Roosts**: Chickens start roosting when they are 8weeks old. Apart from catering for the natural instinct or desire of the chickens to get above the ground at night, perches help materially to keep the bird's feet and plumage clean. Perches can be made from long wooden bars of two squares inches about rounded at the top and flat at the bottom. Fix these parches about 16 inches above the ground and near the walls in such a way that they can be removed for disinfection. Allow a space of 12-inches between two perches. Each bird will need about 8-inches of the perch to roost. The rear perches should rest a little higher than those at the front if they are arranged to be horizontal with the length of the house. This will encourage some of the birds that like to roost high to go to the back perches. Paint the perches occasionally with creosote to prevent insects.
2. **NEST BOXES**: - Each pen of laying birds should be provided with nest boxes for laying eggs. It should be roomy, movable, cool and well ventilated, dark and conveniently located. Nests are usually constructed 14 inches square, 6 inches deep and about 15 inches head allowance. All metal nests are preferred to wood

nests because of easy cleaning and less chance of becoming infested with mites. Empty kerosene tins make excellent boxes. One nest should be provided for every 4 or 5 hens. Dark nests are desirable because they result in less scratching in the nest, less egg breakage and less egg eating. A wooden packing case 18 inches square or a wide mouthed earthen pot can be a suitable nest. Place some sand or soft hay or straw inside. Nests sometimes are also placed inside a run but in that case, care should be taken to prevent crows and other predators by covering the top of the run with wire netting.

3. **TRAP NESTS**: - Each nest is provided with a trap door so that when the poultry attendant releases the hen from the nest, he/she can identify her and mark her leg-band number on the egg. There should be one nest for every three or four birds. Trap nests differ from regular nests in that they are provided with trap doors by which birds shout themselves in when they enter. For the convenience of the poultry attendant, the nests should be placed 18-20 inches above the floor. Trap nests are needed in the poultry houses (Deep litter houses) interested in knowing the performance or breeding of the hens.
4. **FEED HOPPERS**: - The essential features of satisfactory feed hoppers are that they;

- Avoid wastage of feed
- Prevent the birds from getting their feet into the feed and from roosting on the hopper
- Are easy to clean
- Make it easy for the birds to eat from the bottom of the hopper Troughs, pots and pans used for feeding should be of suitable size depending on the age and size of the birds.

1. **Watering Devices**: - An ample supply of water should be made available at all times or egg production is liable to be affected. The water container should contain clean water, kept cool in dry seasons and be easily cleaned because contaminated water tends to spread certain diseases from chicken to chicken. Different designs

of water containers (mostly plastic containers) satisfying the above needs can be provided.

2. **HOVERS**: - This is a heat providing unit. It is made up of pan or Tarpaulin. Brooder unit are maintained with a range of temperature selections for hatchlings. At the warmest, usually the center of the unit, the temperature is maintained at or above 90oF. At the outer edges, the temperature may be as low as 60oF. As the young birds grow; the peak temperature is gradually reduced to about 70oF. Hovers are often used until the birds have reached 4-6weeks of age.

## SELF CHECK 1.2-1

## House equipment installation

TRUE OR FALSE: Write T if the statement is true, if it is false, write F.

1. ________ The poultry house should be equipped with roasts, nests, feed hoppers, water containers and any other items which is essential for satisfactory production.
2. _________ Perches help materially to keep the bird's feet and plumage clean.
3. _________ An ample supply of water should be made available at all times or egg production is liable to be affected.
4. _________ The water container should contain clean water, kept cool in dry seasons and be easily cleaned because contaminated water tends to spread certain diseases from chicken to chicken.
5. _________ Nest is provided with a trap door so that when the poultry attendant releases the hen from the nest, he/she can identify her and mark her leg-band number on the egg.

## ANSWER KEY 1.2-1

## House equipment installation

1. True
2. True
3. True
4. True
5. True

# INFORMATION SHEET 1.2-2

## Prepare and secure bedding materials

**Learning Objective:** After reading this information sheet, you should be able to prepare and secure bedding materials.

**Introduction:**

Litter management has a huge impact on poultry health and comfort. Glenneis Kriel asked Jan Grobbelaar, an independent poultry consultant, for advice on how producers can make the most of poultry bedding.

Poultry litter can either be a farmer's best friend or worst enemy. When managed properly, it will help to prevent diseases and boost farm income by creating a favorable production environment. When managed poorly, it will not only have a negative impact on animal health, welfare and carcass quality, but also on the feed conversion ratio and growth of birds.

Making the most of poultry litter starts with choosing the right kind of bedding material. According to Jan Grobbelaar, training director at Dumela Poultry Solutions in Pretoria, South Africa, bedding material should be light, so that it is easy to manage. It should also be suited for use in compost or animal feed, so that it can be easily be disposed of after use. For health reasons, however, opt for a material that's soft and compressible, with a medium particle size.

"Birds might have trouble walking on material with particles that are larger than 30mm, such as crushed maize cobs, wood chips and wheat straw," says Grobbelaar. "Material with such large particle sizes could hurt their feet and cause conditions such as bumblefoot (ulcerative pododermatitis) or breast lesions. These may have a decimating impact on farm bottom lines, by resulting in carcass downgrades and rejections."

But there are also problems associated with materials that have particle sizes smaller than 2mm – such as sawdust, fine-ground or finely chopped wheat, straw or hay – or that might produce dust, such as wood bark. As Grobbelaar explains, poultry dust – airborne particles of feed and bedding mixed with organic matter from droppings, feathers and dead skin – can affect bird and human health, because, aside from poultry waste, it contains bacteria, viruses and fragments of fungi and spores.

"In the past, it was thought that poultry dust did not have a negative impact on human health, but today we know that it causes respiratory problems in birds and humans," he says, adding that the material used as litter, the age of the litter and climatic conditions all exert an influence on dust levels in a grower house.

An absorbing problem

Another essential quality for material used as bedding is that it should be highly absorbent and quick to dry, to reduce contact between birds and manure. Grobbelaar says pine shavings are still the most popular bedding material in South Africa. Peanut and sunflower hulls are also good, but they have to be turned every week or two to prevent them from caking and to maintain friable conditions. "Fungi that could be damaging to broiler health, will develop if sunflower or peanut hulls become wet," explains Grobbelaar.

Paper and sawdust are not ideal when it comes to moisture absorption, according to Grobbelaar, since paper tends to harden when it gets wet. Sawdust absorbs moisture, but takes too long to release it again – resulting in the litter becoming very wet.

Excess moisture in bedding material is a real problem as it will increase the incidence of breast blisters, skin burns, bruising, condemnations and downgrades. It will also promote bacterial and fungal growth and can cause excessive ammonia emissions. Grobbelaar points out that ammonia levels of about 25 parts per million (ppm) have been associated with poor growth rates, birds' increased susceptibility to Newcastle disease and a build-up of *E. coli*. Prolonged exposure to ammonia levels of 50 to 100 ppm has also been found to cause keratoconjunctivitis and blindness.

The ideal is to maintain litter moisture levels at between 21 and 25 per cent. When the litter exceeds 30 percent, ammonia production will increase as temperatures go up. "To estimate the moisture content of your litter, squeeze a handful of it into a ball," advises Grobbelaar. "If it sticks together in a ball, it is too wet. If it only adheres slightly it will have the proper moisture content. If it doesn't adhere at all, it may be too dry, which is also a problem."

When it comes to the detection of ammonia levels, Grobbelaar said that levels of 10 to 15 ppm can usually be detected by smell. At 25 to 35 ppm it will burn your eyes; at 50 ppm, it could result in broilers exhibiting watery and inflamed eyes, and at 75 ppm, broilers will start showing discomfort and might start jerking their heads.

New bedding material should be stored properly before it is spread in the broiler house, to avoid wet-litter problems. Broiler houses should be managed properly in terms of humidity, ventilation and stocking density and the bedding should be between 70mm and 100mm deep, depending on the type of material used and conditions in the broiler house.

Birds should also receive a high-quality diet to prevent a built up of high moisture levels. Grobbelaar explains that certain dietary ingredients, especially salts, and some drugs cause birds to drink and excrete large amounts of water, which exacerbates wet-litter conditions. In South Africa bedding is renewed after every production cycle.

Keep the heat in

In addition, the ideal bedding material will have low thermal conductivity to retain warmth and act as insulation – it should help to protect broilers from the cold floor. Grobbelaar says this is one of the reasons he does not like sand that much: it has poor thermal conductivity and therefore stays cold. Birds might also struggle to move in sand, if it is spread too deeply.

Finally, your chosen bedding should be free from harmful toxins and contaminants. Grobbelaar says it is best for growers to avoid hardwoods, for example, as these might contain fungi that could be harmful to the broilers.

Cost is usually the biggest driver when it comes to selecting bedding material. But Grobbelaar warns that buying cheap material could actually cost you more in the long run: “While many producers only look at the initial cost, the long-term benefits of using better quality material on animal health and production might make up for the initial difference in price.”

## SELF CHECK 1.2-2

### Prepare and secure bedding materials

## Identification:

1. What absorbs moisture, but takes too long to release it again – resulting in the litter becoming very wet?
2. What is either be a farmer's best friend or worst enemy?
3. Which has a huge impact on poultry health and comfort?
4. What should be free from harmful toxins and contaminants?
5. What should be stored properly before it is spread in the broiler house, to avoid wet-litter problems?

## ANSWER KEY 1.2-2

### Prepare and secure bedding materials

1. Sawdust
2. Poultry litter
3. Litter management
4. Chosen bedding
5. New bedding material

## JOB SHEET 1.2-2

### Prepare and secure bedding materials

**Title: Preparing and securing bedding materials**

**Performance Objective:** At the end of this module the student should be able to prepare and secure chosen bedding materials.

**Supplies/Materials:** PPE, Rice hull, Saw dust , Coco coir and Rice straw

**Steps/Procedures:**

Wear the PPE.
Gather some bedding materials (rice hull, saw dust, coco coir and rice straw
Choose one bedding material which is the best absorbent.
Stored it in a warm and dry place.
Spread it evenly in a poultry house
Do housekeeping

**Assessment Method:**

Demonstration

# PERFORMANCE CHECKLIST 1.2-2

## Prepare and secure bedding materials

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee .. | | |
| 1. Wear the PPE? | | |
| 2. Gather some bedding materials (rice hull, saw dust, coco coir and rice straw | | |
| 3. Choose one bedding material which is the best absorbent? | | |
| 4. Stored it in a warm and dry place? | | |
| 5. Spread it evenly in a poultry house | | |
| 6. Do housekeeping? | | |

## INFORMATION SHEET 1.2-3

### Set-up brooding facility

**Learning Objective:** After reading this information sheet, you should be able to set up brooding facility.

**Introduction:**

How to set up a brooder for baby chicks

Until they are fully feathered, baby chicks require special care. A quality brooder can meet the additional needs of your hatchlings, optimize their growth and ensure good health.

**Setting up your brooder**

There's a lot to consider when setting up a brooder for your baby chicks, but it's worth the effort. Creating a nurturing environment is the difference between a productive flock and birds that never reach their full potential.

- **Container and location.** A few common choices for brooding containers are cardboard, wood and plastic. A cardboard box is the simplest container choice but can be difficult to keep clean and dry. Wooden brooders are more durable but difficult to disinfect after the chicks move outside. Plastic brooding boxes are the easiest to keep clean, but not ideal for air circulation. If you are able to provide supplemental heat in a location free of drafts, building an enclosure in an outbuilding is your best option. No matter where you choose to locate your brooder, it must be able to maintain a temperature of 90 F for the first week.
- **Brooder guards.** If you're able to construct an enclosure for your baby chicks, you'll need a brooder guard to keep them from wandering away from the heat source in the center. You can build yours using cardboard, tar paper or wire. Make it about one foot high and about 18 feet and 10 inches long to form a large circle. These measurements will give you about 3 feet between the brooder and edge of the guard.
- **Bedding.** Once you've chosen a container or built an enclosure, you need to cover the floor of your pen with litter. Use a dry, clean material with good absorption qualities. Wood shavings are most commonly used. However, you need to make sure to select course shavings. Sawdust and fine shavings can lead to litter picking by chicks, which can cause gizzard impaction and lead to death. Chopped wheat straw is another commonly used bedding. It's preferred to other varieties of straws such as rye, oat or barely because they contain oil that reduces their ability to absorb moisture. Other materials to choose from are sugar cane and peanut shells. No matter which material you chose, you should spread it over the bottom of your brooder so that it is at least 4 inches thick. Bedding should be changed at least weekly, but possibly daily depending on the number of chicks you have. The frequency of cleaning will also increase as your chicks grow.

- **Feeders.** When constructing your feeder the most important thing to consider is ease of access. You want to ensure that there is adequate space for each of your chicks to get to the feeder(s) and that the location you've selected is comfortable (not too hot or too cold for the chick). For the first four weeks, you will need to reserve a space measuring 1 inch across and 2 to 3 inches deep per bird at the feeder. From four to eight weeks, your chicks will require a space measuring 2 inches across and 4 inches deep per bird. After eight weeks, they will need a space measuring four inches across and 5 to 6 inches deep per bird.
- **Waterers.** Water is an invaluable part of a chick's nutrition. It's important to make sure water is available for your hatchlings at all times. A 1-quart waterer can serve up to 25 chicks for the first two weeks, but a gallon-sized waterer will be required after that. Placing marbles in the drinking area will attract your chicks to it, prevent them from wading in it and reduce their odds of drowning.
- **Heat sources.** Temperature is one of the most important elements that can affect chick health. Your brooder must be able to maintain a temperature of 90 F through the first week. For every week after, the temperature should be dropped by 5 F until 70 F is reached. Make sure to preheat your brooding area to 90 F a couple days before putting hatchlings in it. There are many ways you can heat your brooder. Some commonly used heat sources for small flocks are electric brooder lamps and liquid propane gas or natural gas brooder stoves.
- **Lighting.** Light intensity should be no less than 20 lux — roughly the amount given off by a street lamp — for the first three days of life. After that just make sure there is enough light for food and water intake, as well as, normal activity. Too much light can cause aggressive behaviors like feather picking or cannibalism.

Cleaning your brooder

You want to make sure you clean your brooder box or area between each set of chicks you raise in it. The process is simple.

1. **Wash.** Use warm, soapy water to wash out your broader box/area and allow it to dry thoroughly.

2. **Sanitize.** Using 1 teaspoon of bleach to every gallon of water, mix a sanitizer to soak your brooder in for 10 minutes. Then rinse and dry.

3. **Repeat.** Follow steps one and two to clean and disinfect feeders, waterers and any other equipment you used in your brooder.

## SELF CHECK 1.2-3

### Set-up brooding facility

**TRUE OR FALSE**

1. A quality brooder can meet the additional needs of your hatchlings, optimize their growth and ensure good health.

2. common choices for brooding containers are cardboard, wood and plastic.

3. There are only two processes involves in cleaning brooder box which includes washing and sanitizing.

4. Temperature is one of the most important elements that can affect chick health.

5. When constructing your feeder the most important thing to consider is ease of access.

# ANSWER KEY 1.2-3

## Set-up brooding facility

1. True
2. True
3. True
4. True
5. True

# JOB SHEET 1.2-3

## Set-up brooding facility

| |
|---|
| **Title: Setting-up brooding facility** |
| **Performance Objective:** At the end of this module the student should be able to set-up brooding facility |
| **Supplies/Materials:** PPE, waterer, water, feeder, feeds, chicks, bedding material. |
| **Steps/Procedures:**<br><br>Wear the PPE.<br>Change the bedding materials<br>wash and sanitize waterer and feeder<br>Pour some clean fresh water into the waterer.<br>Put enough feeds into the feeder<br>Arrange them properly inside the brooding box.<br>Bring chicks inside the brooder box.<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

# PERFORMANCE CHECKLIST 1.2-3

## Set-up brooding facility

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee .. | | |
| 1. Wear the PPE? | | |
| 2 Change the bedding materials with the new one? | | |
| 3.Wash and sanitize waterer and feeder? | | |
| 4. Pour some clean fresh water into the waterer? | | |
| 5. Put enough feeds into the feeder? | | |
| 6. Arrange them properly inside the brooding box? | | |
| 7. Bring chicks inside the brooder box? | | |
| 8. Do housekeeping? | | |

**Learning Outcome #3 FEED CHICKEN**

**Assessment Criteria:**

Suitable **feed materials** are selected based on availability in the locality and nutrient requirements of chicken.

Feed materials are prepared following enterprise prescribed formulation.

Animals are fed based on **feeding managemen**t program.

Feeding is monitored following enterprise procedure.

**Contents:**
Feed materials selection
feeding materials preparation
Feeding management program
Monitoring feeding

**Tools, Materials and Equipment and Facilities**

PPE
Feeding trough
Chopping board
Knife
Plant material

**References:**


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# LEARNING EXPERIENCES/ACTIVITIES

## LO # 3 FEED CHICKEN

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 1.3-1 on Feed materials selection | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.3-1 on Feed materials selection | Try to answer the Self-Check without looking at the information sheet |
| Read the attached Information Sheet 1.3-2 on Feeding materials preparation | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.3-2 on Feeding materials preparation | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.3-2 on Feeding materials preparation | |
| Perform Job Sheet 1.3-2 on Feeding materials preparation | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist 1.3-2 on Feeding materials preparation | |
| Read the attached Information Sheet 1.3-3 on Feeding management program | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.3-3 on Feeding management program | Try to answer the Self-Check without looking at the information sheet |

| | |
|---|---|
| Compare your answer to Answer Key 1.3-3 on Feeding management program | |
| Read the attached Information Sheet 1.3-4 on Monitoring feeding | |
| Answer Self-Check 1.3-4 on Monitoring feeding | |
| Compare your answer to Answer Key 1.3-4 on Monitoring feeding | |

***After doing this activity, you are now ready to proceed to the next module on Grow and harvest chicken.***

# INFORMATION SHEET 1.3-1

## Feed materials selection

**Learning Objective:** After reading this information sheet, you should be able to select materials for feeds

**Introduction:**

The chickens raised in an organic system can eat all the food waste and scraps from your cooking including vegetable peels and stalks, as well as grains. They can also help clean out the pasture in your backyard by eating bugs, grubs, snails, and so on. In an organic farming system, it is a relationship of give-and-take with your chickens. The chicken feed must come from organic sources, for example from certified farms. Organic chicken feed must be grown without using pesticides, antibiotics, herbicides and synthetic fertilizers and without using genetically modified cereals or plants. According to the EU legislation, organic chickens must be born and raised on organic farms and their food must also come from 100% organic sources.Since the number of these farms is not high enough yet, many farmers who make the decision to invest in an organic chicken farm also choose to cultivate their own cereals and forage plants. As feed costs have increased, animal products have become very expensive. If part of the feed could be substituted with root crops such as cassava, then part of the maize ration could be freed for human consumption. The low protein and fibre and high content of soluble carbohydrates (high digestibility) are notable features of the cassava root. Cassava tops, stems and leaves are also available as animal feed and are comparatively high in utilizable protein.

 

**Nutrition requirements for organic chickens**

Chickens and other poultry can qualify as organic only if they are raised under these conditions throughout their **entire life cycle**. Organic chickens have a balanced nutrition based on organic feed, they live in clean housings that provide enough space for movement, have outdoor access and are never treated with antibiotics. Organic chicken farms are inspected by the competent authorities on regular basis to maintain their organic certification.
Agricultural ingredients used to feed organic chickens must come from certified organic farms. The chicken's nutrition should include vitamins, mineral, proteins, amino acids, fatty acids, fiber and energy sources. Some ingredient may be used to replace supplements or additives in foods, for example eggshells and oyster shells may be used as a calcium supplement for egg laying chickens.

The first question is to address the ingredients for homemade chicken feed. Ingredients needed that will provide the right protein, vitamin, and mineral content for the flock are the following.

For basic homemade chicken feed recipe:

- Wheat (hard or soft, winter or spring – it doesn't matter)
- Peas
- Mealworms (live or freeze dried)
- Oats
- Sesame seeds or sunflower seeds

when combined, this recipe yields between 16 – 18% protein – for a growing pullet and a layer, that's the optimum amount of protein.

Both wheat and peas are great for protein (wheat has about 17% protein while the peas are about 24%). The oats are an excellent source of fiber in a homemade recipe, while the sesame and sunflower seeds are great for fat.

There's some controversy about the amount of mealworms a chicken should eat. Given the ability to forage, hens will consume large quantities of bugs – which are almost pure protein.

However: If a chicken eats too much protein, she can develop kidney and other problems. When it comes to mealworms, add a 1/2 cup to their daily ration to start with, and let your chicken tell you if she needs more. If they seem like they need a protein bump, add another 1/2 cup or so of the meal worms.

While I believe it's best to offer live mealworms, not everyone has the time or energy to raise them for a homemade recipe (or the desire, they're bugs after all!). That's okay – Freeze dried ones provide a nice protein bump to your homemade grain too, and they're easier to store.

**Sprout Your Seeds**

This is where the real savings comes in. When you sprout your wheat into fodder, you automatically unlock nutrients, and create a homemade chicken feed that's easier for your flock to digest. In other words, more of the nutrients become bioavailable.

For homemade chicken feed, soak the grains (also known as berries) for 24 hours, then allowing them 3 days to sprout.

You can sprout them longer than 3 days, but you might run into issues with mold. After 3 days, they've started to sprout and unlock the grass, but they haven't turned into a moldy mess that might make your chickens sick.

Once your grain has turned into fodder, you can feed the same weight or volume amount – which ends up being less seed overall.

And the berries have turned into something more nutritious that it could ever be as just a seed. Depending on the type of peas you purchase, you can sprout your peas as well. (Note, if you purchase split peas, you won't be able to sprout fodder).

. **Create a daily ration**

. For 5 chickens, however, in my experience, the following recipe works well for each meal:

- Sprouted seeds (5 cups)
- Peas (2.5 cups)
- Oats (2.5 cups)
- Sesame Seeds (2 tablespoons)
- Mealworms (1/2 cup)

While this homemade recipe usually works well, you might need to scale up or down a bit depending on your flock's needs, and whether you allow them to forage.

**A note about fermenting**

If you want to ferment your homemade chicken feed, you can leave the wheat soaking for another day or so. You will get bubbles what let you know the fermenting is taking place, and the berries will still sprout while submerged.

As with anything fermented, let your nose be your guide – if it smells funny or rancid, toss it. Wheat that's properly fermented will smell something like fresh bread or slightly like beer. I don't recommend letting it soak for longer than an additional 2 days. You will unlock a lot of nutrients as it ferments, but if you wait too long, you can run into other issues.

Make sure you keep your fermenting vessel covered and completely under water. You can ferment the peas as well, following the same steps. Here is my guide to fermenting chicken feed which works for my organic homemade chicken feed recipe or commercial feed.

**Adding supplementary ingredients**

You can add your supplementary ingredients to your homemade chicken feed, such as kelp, garlic, or oregano right before you feed your hens. Just mix them in as you normally would.

I'm a big supporter of giving all three of those supplements to your chickens in a homemade recipe – kelp especially will help ensure your flock gets an iron boost, and the garlic and oregano are great for their antiseptic and immune boosting properties.

This homemade organic chicken feed recipe has been successful for me – I hope it is for you, too!

What to Feed Your Chickens:

- **Commercial Feed:** Commercial poultry pellets should make up the base of your chickens' diet. These pellets are formulated to meet the nutritional needs of chickens and will ensure they're getting everything they need without having to forage. This is especially important for chickens that can't free roam in a large area since food may be limited in a condensed area. Commercial feeds are usually made with foods like sunflower seeds, oats, and wheat.
- **Grasses:** Chickens will eat broad-leaved weeds, like dandelions, and they will eat grasses like clover and Kentucky bluegrass.
- **Insects:** Chickens love to eat bugs and are very effective at helping to control populations of ticks. They will also eat earthworms, beetles, and crickets.
- **Seeds and Grains:** Chickens will eat pumpkin seeds, oats and oatmeal, corn, and cooked rice are all good options to feed your chickens. Just feed these in moderation as they tend to be very nutrient-dense.
- **Grit:** To help digest their food, chickens need to eat grit like sand or coarse dirt. The grit will help the gizzard grind up the food, making it easier to digest and to pull nutrients from.


Image Credit: CClarkPhoto, Shutterstock

**Treats for Chickens:**

- **Vegetables:** Chickens love veggies and will gladly accept whole veggies as well as vegetable peels. Broccoli, cauliflower, zucchini, bell peppers, and tons of other veggies are chicken-safe. Avoid feeding raw potatoes and potato peels, as well as other nightshades, as these can make your chickens sick. Veggies can be fed on a daily basis.
- **Fruits:** Bananas, apple cores and peels, melons, and grapes are good options, as well as other non-citrus fruits. It's best to remove seeds from apple cores prior to feeding since apple seeds contain small amounts of cyanide.
- **Mealworms:** Mealworms are available in freeze-dried and live forms, so you will be able to choose which to feed your chickens. They will happily eat either one, though!
- **Table Scraps:** Chickens will eat just about anything you offer to them. Pancakes, pasta, leftover oatmeal, and unusable scraps from produce like cores and peels. Feed table scraps in moderation and make sure to cut everything up into bite-sized pieces before feeding it to your chickens.
- **Protein:** Chickens are omnivorous, so feeding them meat can be beneficial to their diet. They don't need a lot of meat but will often catch frogs and other small animals when possible as a snack. Chickens also can have some dairy, like cottage cheese, in small quantities. Meat and dairy proteins should be fed in moderation.

**What Not to Feed Your Chickens:**

- Beans
- Raw Potatoes
- Onions
- Citrus
- Candy
- Rhubarb
- Avocado
- Ginger

# SELF CHECK 1.3-1

## Feed materials selection

Read each sentence carefully. Write true if the statement is correct. If not, change the word that makes it wrong.

1. In an organic system, the chicken feed must come from organic sources.
2. If part of the feed could be substituted with root crops such as cassava, then part of the maize ration could be freed for human consumption.
3. The chicken's nutrition should include vitamins, mineral, proteins, amino acids, fatty acids, fiber and energy sources
4. eggshells and oyster shells may be used as a calcium supplement for egg laying chicken.
5. Organic chickens have a balanced nutrition based on organic feed, they live in clean housings that provide enough space for movement, have outdoor access and are never treated with antibiotics.

# ANSWER KEY 1.3-1

## Feed materials selection

1. True
2. True
3. True
4. True
5. True

# INFORMATION SHEET 1.3-2
# Feeding materials preparation

**Learning Objective:** After reading this information sheet, you should be able to prepare feeding materials.

**Introduction:**

Making your own chicken feed is a great way to save money and allows you to know exactly what you're feeding your chickens. If you want to feed your chickens organically, use organic ingredients in these recipes. Try the chicken feed recipe for laying hens, or make the broiler feed if you are raising broiler hens. Both recipes are rich in protein and nutrients and will help to nourish your chickens.

**Ingredients**

**Making Chicken Feed for Laying Hens**

- 107 pounds (49 kg) of whole maize meal
- 41 pounds (19 kg) of soya
- 28 pounds (13 kg) of fish meal
- 31 pounds (14 kg) of maize bran
- 13 pounds (5.9 kg) of limestone powder

*Makes 220 pounds (100 kg) of chicken feed*

**Creating Feed for Broilers**

- 250 pounds (110 kg) of cracked corn
- 150 pounds (68 kg) of ground roasted soybeans
- 25 pounds (11 kg) of rolled oats
- 25 pounds (11 kg) of alfalfa meal
- 25 pounds (11 kg) of fish or bone meal

- 10 pounds (4.5 kg) of aragonite (calcium powder)
- 15 pounds (6.8 kg) of poultry nutri-balancer

*Makes 500 pounds (230 kg) of chicken feed*

**Making Chicken Feed for Laying Hens**

1. **Measure the ingredients into a container.** Add 107 pounds (49 kg) of whole maize meal, 41 pounds (19 kg) of soya, 28 pounds (13 kg) of fish meal, 31 pounds (14 kg) of maize bran, and 13 pounds (5.9 kg) of limestone powder into a container. This recipe makes 220 pounds (100 kg) of chicken feed, so you will need a large bucket or barrel to mix and store the feed in.[1]
    - Use organic ingredients if you want to make the chicken feed organic.
    - Purchase the ingredients from a bulk goods store or a farm shop.

**2.Mix the ingredients until they are thoroughly combined.** Stir the feed with a shovel until all the ingredients are evenly dispersed throughout the container. This ensures that the chickens will receive the nutrients from the different ingredients when they are fed.[2]

- Make sure that you mix the ingredients that are in the bottom of the container.
- This may take a few minutes if you have made a large batch. Allow 2-3 minutes to mix a large bucket.
- If you have made a very large batch of chicken feed, use a spade to mix the ingredients.

**3.Give each chicken 0.28 pounds (0.13 kg) of feed per day.** Multiply the feed needed per chicken by the number of chickens that you have. For example, 6 chickens x 0.28 pounds (0.13 kg) = 1.68 pounds (0.76 kg) of feed in total. Place the food into a feeder or sprinkle it on the ground in front of them.

- If you are using a feeder, simply pour the feed into the hole at the top and let it trickle down into the feeding plate. Purchase a feeder from a farm store or make your own chicken feeder.

**4.Store the chicken feed in a cool, dry place for up to 6 months.** Garages or barns are ideal places to store chicken feed. Check the feed for mice, bugs, and mould before you give it to the chickens. If the feed has been contaminated, it is safest to throw it away.[3]

- If you don't have a shed to store the feed in, put a lid on the container and keep it out of direct sunlight.

**Creating Feed for Broilers**

**1**

**Mix the cracked corn and ground roasted soybeans in a container.** Measure 250 pounds (110 kg) of cracked corn and 150 pounds (68 kg) of ground roasted soybeans into a large container, such as a barrel or feed container. Mix the ingredients with a shovel until they are thoroughly combined.[4]

- Choose a container that has a lid. This will make it easier to store the feed.
- If you don't have a big enough container, halve the recipe.
- This feed works well for broiler chickens as it has lots of protein to help the chickens grow.
- Use organic ingredients if you want to make organic feed.

**2**

**Stir the rolled oats, alfalfa meal, and fish or bone meal into the mixture.** Measure 25 pounds (11 kg) of rolled oats, 25 pounds (11 kg) of alfalfa meal, and 25 pounds (11 kg) of fish or bone meal into the container. Mix the ingredients into the cracked corn and soybeans until all the ingredients are evenly distributed in the container.[5]

- Purchase the ingredients from a farm store or a bulk foods shop.

**3**

**Add the aragonite and poultry nutri-balancer to the container.** Measure 10 pounds (4.5 kg) of aragonite (calcium powder) and 15 pounds (6.8 kg) of poultry nutri-balancer into the feed. Mix the ingredients thoroughly so that the powders are thoroughly distributed through the feed. The poultry nutri-balancer is an important addition to the feed, as it ensures that the chickens receive the nutrients that they need to grow quickly.[6]

- If you can’t find these ingredients at your local farm shop, look online or ask your vet to recommend a distributor.
- Aragonite is mineral found in limestone and is a great source of calcium.

**4**

**Feed each chicken 0.6 pounds (0.27 kg) of the mixture each day.** Multiply the amount of feed per chicken by the number of chickens in the coop. Place the feed into a feeder or throw it on the ground once per day.

- Use 3 pounds (1.4 kg) of feed for every 5 chickens.
- It is important not to over-feed this mixture to cornish boilers as this can cause fatal heart attacks. This is uncommon as the chickens do not tend to eat more food than they need.

**5**

**Store the chicken feed in a covered container for up to 6 months.** Place a lid over the container of feed and place it in a cool and dry area, such as a garage or barn. This will help to stop the feed from going mouldy or being contaminated by bugs.[7]

- If you see any signs of mice or bugs in the feed, it is best to throw it away and make a new batch.

## SELF CHECK 1.3-2
## Feeding materials preparation

**Write TRUE if the statement is correct, write false if it is incorrect**.

1. Making your own chicken feed is a great way to save money.
2. If you want to feed your chickens organically, use organic ingredients in these recipes.
3. The ingredients for laying hen feeds are different from the broiler feeds.
4. Store the chicken feed in a covered container for up to 6 months.
5. Garages or barns are ideal places to store chicken feed. Check the feed for mice, bugs, and mould before you give it to the chickens.

# ANSWER KEY1.3-2

## Feeding materials preparation

1. True
2. True
3. True
4. True
5. True

# JOB SHEET 1.3-2
# Feeding materials preparation

| |
|---|
| **Title: Preparing feeding materials** |
| **Performance Objective:** At the end of this module you should be able to prepare feeding materials. |
| **Supplies/Materials:** PPE, whole maize meal soya<br>fish meal maize bran limestone powder |
| **Steps/Procedures:**<br><br>wear the PPE.<br>Measure the ingredients into a container<br>Mix the ingredients until they are thoroughly combined<br>Give each chicken 0.28 pounds (0.13 kg) of feed per day<br>Store the chicken feed in a cool, dry place for up to 6 months.<br>Do housekeeping. |
| **Assessment method:**<br>**Demonstration:** |

# PERFORMANCE CHECKLIST 1.3-2

## Feeding materials preparation

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee .. | | |
| wear the PPE? | | |
| Measure the ingredients into a container | | |
| Mix the ingredients until they are thoroughly combined? | | |
| Give each chicken 0.28 pounds (0.13 kg) of feed per day? | | |
| Store the chicken feed in a cool, dry place for up to 6 months? | | |
| Do housekeeping? | | |

# INFORMATION SHEET 1.3-3
# Feeding management program

**Learning Objective:** After reading this information sheet, you should be able to differentiate different feeding program.

**Introduction:**

Chickens are great additions to both traditional and urban farms. They help with pest control and produce enough eggs that many people sell them or give them away. Chickens are also foragers by nature, so they will happily spend the entire day pecking around for something to eat. There are a ton of food options for chickens, allowing you to provide them with a healthy diet and fun treats as well. Here are the things you need to know about feeding your chickens!

**How Often to Feed Chickens**

Ideally, you should split your chicken's feed into two servings daily. If you're home during the day, you can even make this 3-4 small feedings. Chickens enjoy small, frequent meals as opposed to large meals once a day. It's best to feed your chickens their pellets in a feed trough of some sort for easy cleaning but treats and scraps can be tossed on the ground to provide an enriching hide-and-seek game for your chickens. Just make sure you're not overfeeding or you may end up with leftover rotting food.

A major benefit of feeding small meals twice daily is that it decreases the risk of attracting pests from food left sitting in the feed trough. Pick up any unfinished food at night to avoid attracting mice, possums, and other pest animals.

**The Importance of Water in a Chicken's Diet**

Basically, all living things require water in some form for survival, and chickens are no different. They should always have access to clean water to prevent dehydration. A single chicken can drink up to a liter of water daily, and sometimes will drink even more during hot weather. Take this into account when you're filling up your chickens' waterer, and make sure to account for environmental factors like evaporation.

**What Additional Supplements Do Chickens Need?**

Sometimes, chickens are not able to fully absorb and make use of the nutrients in their food, so supplementation may be necessary. Not all supplements are needed all the time, but here are some supplementation ideas for you to keep your chickens healthy.
They should be provided with a source of grit, especially if they are not free range. Free range chickens often pick up gravel and dirt as they roam, fulfilling their grit needs. Chickens also need adequate calcium for egg production, and this can be achieved by feeding them dried eggshells that have been crushed or ground into powder. There are also oyster shell supplements available at most feed stores.
During the summer, adding electrolyte supplements into the water may be necessary and powdered electrolytes are usually available at feed stores. Powdered probiotics can be added to your chickens' food to maintain digestive health. Apple cider vinegar may help thin mucus and garlic's antibacterial properties can help reduce the risk of illness, although it may alter the taste of your eggs.

**Ad libitum feeding versus food restriction**

Ad libitum feeding means that the diet is available at all times.
Restricted feeding refers to restricting the amount of food while still ensuring nutritional adequacy.

# SELF CHECK 1.3-3
# Feeding management program

**True or False: Write true if the statement is correct, write false if it is incorrect.**

1. Chickens are also foragers by nature, so they will happily spend the entire day pecking around for something to eat.
2. Chickens enjoy small, frequent meals as opposed to large meals once a day
3. Chickens also need adequate calcium for egg production, and this can be achieved by feeding them dried eggshells that have been crushed or ground into powder.
4. Ad libitum feeding means that the diet is available at all times.
5. Restricted feeding refers to restricting the amount of food while still ensuring nutritional adequacy.

# ANSWER KEY 1.3-3

## Feeding management program

1. True
2. True
3. True
4. True
5. True

# INFORMATION SHEET 1.3-4
# Monitoring feeding

**Learning Objective:** After reading this information sheet, you should be able to monitor feeding.

**Introduction:**

Most of the businessmen and farmers use traditional poultry farming methods. The traditional poultry farms lack proper and effective management to maintain health and growth of chicken. All the poultry activities like filling the water tank, monitoring temperature, time to time feeding of chicken, cleaning the chicken waste and light control in the farm are done manually. Hence, a large manpower is required. So without automation it requires manual work to take care of poultry (chickens) and what if the care taker/owner is not present in farm and due to some environmental conditions, the poultry birds get harms or may die, that may affect the business. Many problems arise while taking good care of the poultry birds as itis a very tedious and intricate task which demands lot of alertness and minimum errors. These sensitive creatures are prone to lot of diseases which might be a hindrance in the business

**What to Do if Your Chicken Isn't Eating**

If you have a chicken that isn't eating, you can offer things like a mush of commercial feed mixed with warm milk or water. Sometimes, all your chicken need is to be handfed for a little while, so you can try this as well as attempting feeding via syringe or spoon.

If your chicken continues to refuse to eat or if you have multiple chickens who are suddenly showing inappetence, you should contact your veterinarian immediately. It’s possible your chickens may have encountered toxins or poisons and need veterinary care. When in doubt, contact your vet! Veterinarians are happy to answer questions, and many would rather see your chicken before they are very ill. This gives your chicken the best chance of regaining health.

# SELF CHECK 1.3-4

## Monitoring feeding

**True or false: Write T if the statement is true, write F if it is false.**

1. If you have a chicken that isn't eating, you can offer things like a mush of commercial feed mixed with warm milk or water.
2. Sometimes, all your chicken need is to be handfed for a little while
3. All the poultry activities like filling the water tank, monitoring temperature, time to time feeding of chicken, cleaning the chicken waste and light control in the farm are done manually,
4. If your chicken continues to refuse to eat or if you have multiple chickens who are suddenly showing inappetence, you should contact your veterinarian immediately.
5. Chicken are prone to lot of diseases which might be a hindrance in the business.

# ANSWER KEY 1.3-4

## Monitoring feeding

1. T
2. T
3. T
4. T
5. T

## Learning Outcome #4 Grow and harvest chicken

**Assessment Criteria:**

Growth rate is monitored based on enterprise procedures

Health care program are implemented based on enterprise procedures

Sanitation and cleanliness program are implemented based on enterprise procedure

Organic waste for fertilizer formulation are collected.

Suitable chicken for harvest are selected based on market specifications.

Production record is accomplished according to enterprise procedure.

**Contents:**

Monitor growth rate
Healthcare program
Sanitation and cleanliness program
Organic waste collection
Suitable chicken for harvest selection
Production record

**Tools, Materials and Equipment and Facilities**

Poultry farm

Fully grown broilers

Paper

Pen

Calculator

Weighing scale

**References**

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## LEARNING EXPERIENCES/ACTIVITIES

### LO # 4 Grow and harvest chicken

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 1.4-1 on Monitor growth rate | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.4-1 on Monitor growth rate | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-1 on Monitor growth rate | |
| Read the attached Information Sheet 1.4-2 on Healthcare program | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.4-2 on Healthcare program | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-2 on Healthcare program | |
| Read the attached Information Sheet 1.4-3 on Sanitation and cleanliness program | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.4-3 on Sanitation and cleanliness program | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-3 on Sanitation and cleanliness program | |
| Read the attached Information Sheet 1.4-4 on Organic waste collection | You can ask the assistance of your trainer explain further the topics you cannot understand |

| | |
|---|---|
| Answer Self-Check 1.4-4 on Organic waste collection | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-4 on Organic waste collection | |
| Read the attached Information Sheet 1.4-5 on Suitable chicken for harvest selection | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.4-5 on Suitable chicken for harvest selection | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-5 on Suitable chicken for harvest selection | |
| Read the attached Information Sheet 1.4-6 on Production record | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 1.4-6 on Production record | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 1.4-6 on Production record | |

# INFORMATION SHEET 1.4-1
# Monitor growth rate

**Learning Objective:** After reading this information sheet, you should be able to monitor growth rate of a broiler

**Introduction:**

**Management Minute – Broiler Weight Monitoring**

Broiler weight is an important parameter when growing broilers. After all, the end goal of raising broilers is to get as many birds to market as possible, using the least amount of feed, and producing the greatest amount of meat. With pressures on the industry to reduce antibiotic usage, there is a greater focus on finding new ways to maximize birds' growth, health, and well being from the earliest stages.

It has long been known that the better start a broiler bird gets the better chance for that bird to reach its genetic potential. Broiler chickens have the genetic potential for significant weight gain over a very short period of time. During the first 7 days, 80% of the bird's energy is used for growth and only 20% for maintenance, indicating the importance of this period in the chicken's life. Weighing around 42g at hatch, broilers can achieve a weight of 2,800g (6 lbs.) within the next 42 days – an average daily growth rate of 66g (.146 lbs.). This growth rate is particularly significant within the first seven days, as the bird has the potential to increase its body weight by 450% from day zero to day seven. Furthermore, according to management guidelines for the modern broiler, the birds are supposed to increase their live weight 4.25 times during the first 7 days, from approximately 40g to 180g. Seven-day weight of the Cobb 500 in comparison with hatch weight has increased by 300% over the last 20 years.

Good early development of the chick reaching high 7-day weights will have a significant positive impact on the bird for the rest of its life. It will improve the nutritional maturity of the bird and accelerate development of the gastro-intestinal tract. Muscle growth and development of muscle morphology improves, as will long-term positive metabolic effects. Below are performance numbers published by Ross and Cobb as to what should be expected for weight gains on their straight run broilers.

ROSS 308

| Day of Age | Body Weight-g (lbs.) | Average Daily Gain/Week- g (lbs.) |
|---|---|---|
| 0 | 43 | |
| 7 | 208 (0.459) | 23.50 (0.052) |
| 14 | 519 (1.14) | 44.46 (0.098) |
| 21 | 985 (2.17) | 66.56 (0.147) |
| 28 | 1573 (3.47) | 84.07 (0.185) |
| 35 | 2235 (4.93) | 94.47 (0.208) |
| 42 | 2918 (6.43) | 97.67 (0.215) |
| 49 | 3583 (8.49) | 95.04 (0.21) |
| 56 | 4203 (9.26) | 88.47 (0.195) |

Cobb 500

| | | |
|---|---|---|
| 0 | 42 | |
| 7 | 193 (0.425) | 21.50 (0.047) |
| 14 | 528 (1.164) | 47.86 (0.106) |
| 21 | 1018 (2.24) | 70.00 (0.154) |
| 28 | 1615 (3.56) | 85.28 (0.188) |
| 35 | 2273 (5.01) | 94.00 (0.207) |
| 42 | 2952 (6.51) | 97.00 (0.214) |
| 49 | 3617 (7.97) | 95.00 (0.209) |
| 56 | 4227 (9.32) | 87.14 (0.192) |

As important as bird weight is in the broiler industry, it has never been a common practice to routinely weigh broilers, mainly due to the physical time it takes to weigh birds on a multiple barn farm. With the development of accurate automatic scales that can have multiple platforms in a single barn, collecting weights on a broiler farm has become easier, whether you raise straight run birds or sexed birds. Actively weighing your birds gives vital data to help better manage birds to reach their genetic potential. Weighing broilers can confirm how effective different management practices are. Why spend time and energy doing something that does not give a payback in performance?

Every management technique implemented should be to maximize performance of the flock. There are weighing systems available with computer software that will collect and calculate important data such as average daily weight gains, daily actual weight, coefficient of variation, uniformity, and number of birds weighed. Some programs have built in breed weight curves to be able to easily compare gains with documented weight curves for specific breeds. These programs can have mortality inputted to be able to track total mortality easily. Some of these programs will create graphs, email information, and send information to Excel spreadsheets which allows past flocks to be stored for easy comparison.

One of the best technological advancements for these weighing systems is the ability to be monitored through a controller or the internet. These systems allow the user to access their information from an offsite location over the internet or just view and collect the data at a central location on the farm. A production data analysis can help determine which flocks are meeting their genetic potential and which flocks are not, as well as which management practices are beneficial.

As weights are collected, if the weights of the flock are not following the recommended breed guidelines, an investigation can begin to determine the cause, instead of waiting until the flock ships only to learn the flock has below average weight gain. Some of the problems that could be discovered to cause poor weight gain include disease challenges, temperature or air quality issues, inadequate feed space for all birds to eat (birds not evenly spaced throughout barn), slow feed delivery or distribution, and feed formulation or quality problems. By catching a problem early, it is possible to get the birds back on track, maybe not completely recover but at least work to get the birds back on their growth curve and maximize remaining growth days.

The success of any poultry program must be driven with data. Good decision-making is dependent upon the quality of data and ensuring that the numbers are accurate and complete. The level of uniformity largely contributes to the final result, and as with any business, increasing profits with positive final results is what one is after. The broiler industry is no different. Invest in a tool to maximize performance and profits.

. Overall
2. Specific
1. Overall
In this case you need to weigh the Day Old Chick and record the feed eaten since day 1. Say you harvested at 50 days. Then the growth rate is the weight increase in the 50 days minus the weight of the day old chick. Then the growth rate is the increase in weight over the 50 days divided by 50 days.
2. Specific
Say you want to know the data from 31 days to 50 days. Record the weight of your poultry at 31 days. Start recording the feed consumed from day 30 until day 50.Total feed consumed is the cumulative feed from day 31 to day 50, say this is X. The weight increase of the poultry is the weight at 50 days minus the weight at 31 day, say Y. FCR = X divided by Y. Growth rate from 31 days to 50 days is the weight increase divided by 20 days!
Cheers
Growth rate means average gain in body weight for prescribed period.
Feed conversion efficiency or Feed conversion ratio means total feed consumed divided by total gain in weight for a particular periods.
It is very easy
Growth can be taken in growing birds only.
It can be calculated that total weight of eggs production per kg of feed during particular period of time.
This is the feed conversion efficiency.
do all calculation on daily bases. so you have a history of the flock (Broilers) then calculate FCR, and the Gain also on Daily bases then fit an equation to the data you have for the whole period. this will help you estimate what you need if you are using a simulation model, or comparing data between to flocks. the important thing is you need also to record the Temperature inside the shed and relative humidity as well. because this may effect data form season to season. also Note that $CO_2$ will be higher at the beginning of the run comparing to at the end of the run NH3 on the other hand will act oppositely, because of accumulated manure on the litter. you need to have all these parameters in the acceptable range or level to be sure your data representing the flock. and to do that you need to measure the ventilation rate to control $CO_2$ and $NH_3$ as well as Temperature.

# SELF CHECK 1.4-1

## Monitor growth rate

Fill in the blanks. Write your answer on the space provided.

1. What is an important parameter when growing broilers?
2. Which chicken type has the genetic potential for significant weight gain over a very short period of time?
3. What can only confirm how effective different management practices are?
4. What can be taken in growing birds only?
5. Which means average gain in body weight for prescribed period?

# ANSWER KEY 1.4-1

## Monitor growth rate

1. Broiler weight
2. Broiler chickens
3. Weighing broilers
4. Growth
5. Growth rate

## INFORMATION SHEET 1.4-2

## Healthcare program

**Learning Objective:** After reading this information sheet, you should be able to appreciate healthcare program.

**Introduction:**

**Health Management**

The best fed and housed stock with the best genetic potential will not grow and produce efficiently if they become diseased or infested with parasites. Therefore good **poultry health management** is an important component of poultry production. Infectious disease causing agents will spread through a flock very quickly because of the high stocking densities of commercially housed poultry.

For poultry health management to be effective a primary aim must be to prevent the onset of disease or parasites, to recognise at an early stage the presence of disease or parasites, and to treat all flocks that are diseased or infested with parasites as soon as possible and before they develop into a serious condition or spread to other flocks. To be able to do this it is necessary to know how to recognise that the birds are diseased, the action required for preventing or minimising disease and how to monitor for signs that the prevention program is working.

**Principles Of Health Management**

The key principles of poultry health management are:

1. Prevention of disease
2. Early recognition of disease
3. Early treatment of disease

As much as is possible disease should be prevented. It is easier and less damaging to prevent disease than it is to treat it. However, it must not be assumed that all disease can be prevented. Inevitably, some will get past the defences, in which case it becomes imperative that the condition is recognised as early as possible to allow treatment or other appropriate action to be implemented as soon as possible to bring the situation under control to limit damage to the flock.

Disease


**Avian influenza causing depression Source: CSIRO**

A disease is any condition that interferes with the normal functioning of the cells, tissues, organs and the whole body systems. Diseases of poultry have many causes and include:

1. Deficiencies of essential nutrients such as vitamins, minerals or other nutrients.
2. The consumption of toxic substances such as poisons.
3. Physical damage e.g. environmental extremes and injury.
4. Internal and external parasite infestations such as lice and worms.
5. Infectious disease caused by micro-organisms such as bacteria and viruses.

Diseases that result from nutrient deficiencies, consumption of toxic substances and physical damage are referred to as non-infectious diseases. These diseases cannot be passed from bird to bird and members of the flock must share a common experience for individuals to contract these non-infectious diseases. In the widest sense, infectious diseases are caused by microorganisms that include parasites, fungi, protozoa, bacteria, mycoplasmas, chlamydia and viruses. These diseases are often also called contagious diseases meaning that they can be passed from one bird to another either directly or indirectly.

**Direct transmission** occurs when one diseased bird passes the cause of the disease via direct contact to a susceptible healthy bird. Such passage may be horizontal transmission (from one bird to another) or vertical transmission (from parent to offspring) via the egg or sperm either inside the egg or on the shell. **Indirect transmission** occurs when the causal organism is passed from one bird to another via an intermediate host such as insects, earthworms, snails or slugs, wild birds or animals or some other object such as equipment, food or water, vehicles, people, respiratory droplets, litter or faeces.

**Causes Of Infectious Disease**

Organisms and microorganisms that have the potential to cause harm, such as disease in animals, are called pathogens or disease vectors. There are many different types of pathogens that may be transferred from one bird to another or from one flock to another by many different means. These pathogen types include:

- Viruses
- Bacteria
- Fungi
- Protozoa
- Internal parasites
- External parasites

# SELF CHECK 1.4-2
# Healthcare program

True or False: Write T if the statement is correct if it is incorrect, write F.

1. The best fed and housed stock with the best genetic potential will not grow and produce efficiently if they become diseased or infested with parasites.
2. Infectious disease causing agents will spread through a flock very quickly.
3. Direct transmission occurs when one diseased bird passes the cause of the disease via direct contact to a susceptible healthy bird.
4. Indirect transmission occurs when the causal organism is passed from one bird to another via an intermediate host such as insects, earthworms, snails or slugs, wild birds or animals or some other object such as equipment, food or water, vehicles, people, respiratory droplets, litter or faeces.
5. Diseases that result from nutrient deficiencies, consumption of toxic substances and physical damage are referred to as non-infectious diseases.

# ANSWER KEY 1.4-2

## Healthcare program

1. T
2. T
3. T
4. T
5. T

# INFORMATION SHEET 1.4-3
# Sanitation and cleanliness program

**Learning Objective:** After reading this information sheet, you should be able to appreciate the importance of sanitation and cleanliness program.

**Introduction:**

**Sanitation, Cleaning, and Disinfecting Poultry Facilities**

Diseases and infections have always been a major concern to the poultry industry. Fortunately, microbial contamination can be prevented and controlled using proper management practices and modern health products.

**Microorganisms are everywhere!** Some are relatively harmless, while others can cause disease. Some pose a lethal threat to one species of animal while remaining harmless to another species. Some organisms are easily destroyed, while others are very difficult to eliminate. The moral is, **"Treat all microorganisms as if they are a severe threat to the chick's livelihood."**

Three terms are commonly used to describe microbial control:

* **Sterilization –** Destroying all infective and reproductive forms of all microorganisms (bacteria, fungi, virus, and the like).
* **Disinfection –** Destroying all vegetative forms of microorganisms. Spores are not destroyed.
* **Sanitation –** Pathogenic organisms are present but are not a threat to the birds' health.

Many producers have the impression that they create a "sterile" condition because they use disinfectants, when they may only achieve a sanitized condition at the very best.

The most important thing to remember when striving for a sanitized environment is that **cleanliness is essential**. Proper cleaning removes most germs and is always done before using disinfectants. This applies to all areas, including floors, walls, equipment, and personnel.

It is extremely important to remove as much organic matter as possible from surfaces being disinfected. After removing dust, chick down, droppings, tissue residues, and such, thoroughly clean surfaces, using warm water and appropriate cleaning aids. Focus on selecting the proper detergent to produce the cleanest environment possible with variations in water hardness, salinity, and pH. A thorough rinsing with enough clean, sanitized water completes the cleaning process and removes most lingering residues of detergents, organic matter, or microbial germs.

Only after facilities are thoroughly clean do you treat surfaces with an appropriate disinfectant solution. Not all disinfectants are suited for every situation.

When selecting the disinfectant, carefully consider these:

* The type of surface being treated.
* The cleanliness of the surface.
* The type of organisms being treated.
* The durability of the equipment/surface material.
* Time limitations on treatment duration.
* Residual activity requirements.

If the surface is free of organic matter and residual activity is not required, quaternary ammonium compounds or halogen compounds can be used effectively. However, if surfaces are difficult to clean, residual activity is required, or the contaminating organisms are difficult to destroy, then multiple phenols or coal tar distillates may be needed.

Be careful that the disinfectant, when used as directed, meets your requirements. Be reasonable and don't expect the product to produce impossible results.Otherwise, select a different product or change disease control practices.

Although many disinfectants are available, the disinfectant you select must be effective for the conditions being used.

Here are several considerations for getting the best results from a disinfectant:

* Consider the disinfectant's effectiveness on organisms of greatest concern.

Not all disinfectants are
effective against all organisms.

* Clean and disinfect in separate operations.
* Disinfectant solutions are more effective when applied as warm solutions rather than cold solutions. Hot solutions can reduce disinfectant efficiency.
* Few disinfectants are effective instantaneously; allow enough contact time (usually 30 minutes is sufficient).
* Embryos are very sensitive and severely affected by chemical vapors. Use disinfectants having least effect on embryo development.
* Allow all surfaces to dry thoroughly before reuse. Dryness reduces reproduction and spread and transport of germs.
* Improper use of disinfectants can damage or hinder the function of equipment. Some disinfectants are corrosive or clog spray nozzles of water systems.
* Always follow label directions for their safe use. Never sacrifice personal safety for cost savings or productive efficiency.

Disease-free surfaces can be compromised if you do not properly maintain facilities. You can unknowingly act as a germ carrier and become a major source of infection. Provisions must be available for frequent washing of hands and footwear. Freshly laundered clothing and caps can significantly reduce the spread of germs. Restricted movement of personnel within specific areas also reduces the distribution of organisms.

The risk posed by disease causing organisms is a constant challenge. Use effective control measures rather than trusting visual cleanliness as an indicator of sanitation. A surface that looks clean is not necessarily disease-free. Assuming so may be fatal to the birds and management program.

| Disinfectant Classifications | | |
|---|---|---|
| **Disinfectant Type** | **Recommended Use** | **Considerations** |
| **Alcohols** | Small utensils | Poor residual activity, fire hazard, expensive |

| | | |
|---|---|---|
| **Halogens** | Water sytems, foot baths | Corrosive, poor residual activity, ineffective in presence of organic material |
| **Quaternary ammonias** | Incubation equipment, feeding systems | Non-corrosive, non-irritating, limited residual activity and effectiveness with organic matter |
| **Phenols** | General house use | Slightly irritating, good residual activity, effective with organic matter |
| **Coal tar distillates** | General house use | Coorosive, irritating, good residual activity, effective with organic matter |
| **Aldehydes** | Fumigating incubators/eggs | Highly toxic, slight residual activity, sporicidal, fungicidal |
| **Oxidizing agents** | Small utensils | Poor residual activity, corrosive, innefective in presence of organic material |

# SELF CHECK 1.4-3
# Sanitation and cleanliness program

Write YES if the statement is correct write NO if it is in correct.

1. Microbial contamination can be prevented and controlled using proper management practices and modern health products.
2. Sterilization – Destroying all infective and reproductive forms of all microorganisms (bacteria, fungi,
3. Proper cleaning removes most germs and is always done before using disinfectants
4. The most important thing to remember when striving for a sanitized environment is that cleanliness is essential
5. Focus on selecting the proper detergent to produce the cleanest environment possible with variations in water hardness, salinity, and pH. A thorough rinsing with enough clean, sanitized water completes the cleaning process and removes most lingering residues of detergents, organic matter, or microbial germs.

# ANSWER KEY 1.4-3
# Sanitation and cleanliness program

1. YES
2. YES
3. YES
4. YES
5. YES

# INFORMATION SHEET 1.4-4

## Organic waste collection

**Learning Objective:** After reading this information sheet, you should be able to collect organic waste.

**Introduction:**

**Organic waste**, or biodegradable waste, is a natural refuse type that comes from plants or animals. It comes in manifold forms – biodegradable plastics, food waste, green waste, paper waste, manure.

### Organic Waste Recycling

Thankfully, a landfill doesn't have to be the end of the road for organic materials. For a disposal solution that is advantageous and eco-friendly (rather than wasteful and damaging), we can borrow one of nature's simplest solutions: composting.

With time, all organic refuse will decompose. Composting is the controlled, accelerated process of recycling those decomposed organic materials into a nutrient-rich soil. Small amounts of organic waste can be composted in your backward. It has been estimated that you could *remove* as much as 500 lbs of organic material from your home each year.

On a larger scale, businesses also have the option of sending their organic waste to special facilities, where it will be recycled into usable resources. If your business generates this waste type and would benefit from a more efficient management program.

### Types of poultry waste

1. Poultry manure
2. Hatchery waste
3. Slaughter house and processing plant waste
4. Dead bird

## 1. POULTRY MANURE

Two main types of waste are produced by poultry enterprises depending on the rearing system adopted on the farm

• Poultry litter – Waste from deep litter systems
• Cage layer waste – Excreta collected under the cages, spilled feed and feathers.

### a. Drying

• Oldest, cheapest and feasible method
• Dried under sunlight and depends on lengths of time, climate and humidity.
• Drying the manure with heat results in loss of energy and nitrogen.
• Thin bed drying prevents the breeding of flies, reduces obnoxious odours and maintains the nutrient value of the manure particles.
• The faster the manure is dried, the higher is the nitrogen value.

### b. Heaping

• Deep stacking of poultry waste produces considerable heat and had been shown to destroy coliforms.
• The maximum temperature was reportedly attained in 4-8 days.

### c. Poultry manure as organic fertilizer

• Poultry manure applications increase the moisture holding capacity of the soil
• Improve lateral water movement, improves irrigation efficiency and decreases drought
• Improve soil retention and uptake of plant nutrients.
• Increase the number and diversity of soil microorganisms.

### d. Biogas / Electricity generation from poultry litter

• Poultry litter has a good calorific value for power generation by combustion under controlled conditions.
• The technology for anaerobic conversion of poultry manure to biogas (methane) has been developed.
• Electricity production facilities estimated assuming poultry litter utilization rates of 1000 tons/year, 10,000 tons/year, and 50,000 tons/year for various technologies range from 34–70 kW, 340–700 kW, and 1.7–3.5 MW, respectively.
• Economic analysis accounting for capital expenditures, operation and maintenance costs, litter cleanout and transportation, and recoverable

sludge/ash value reveal that gasification at a small scale (100 kW) and medium scale (1 MW) is potentially economically viable compared to anaerobic digestion and combustion.

**e. Composting**

- Can be stored for long time
- Aerobic bacterial action occurs
- The top foot is composed of fresh manure, the bottom foot is in an anaerobic condition and the central portion is undergoing composting.
- The essential requirement in managing the deep pit is that the fresh, wet material be adequately aerated to remove the moisture.
- To further the composting process and to prevent odours the pit must be watertight so that seepage water cannot enter.
- Little or no odour arising from the pits and manure removal may be delayed for years.

**f. Pond disposal**

- Fresh poultry manure may be flushed into an open, shallow pond.
- Bacterial action reduces the waste material to a smaller volume.
- Bacterial growth occurs only during the warm months, the use of ponds is seasonal.
- The resulting solution may be spread in its liquid state on farmland.
- Aerobic action produces little odour as the sludge builds up, anaerobic activity takes place and odours may be pronounced.

**g. Aeration**

- Water is poured into the trough to keep the manure fluid and pumps keep the sludge circulating. The effluent is aerated by paddles.
- The addition of oxygen by the paddles increases the activity of aerobic bacteria, greatly reducing the incidence of any odours.
- The material is removed in liquid form and usually spread on the land. The material is practically odourless.

**2. HATCHERY WASTE DISPOSAL**

- Solid hatchery waste comprises empty shells, infertile eggs, dead embryos, late hatchings and dead chickens and a viscous liquid from eggs and decaying tissue.
- Wastewater comes from water used to wash down incubators, hatchers and

chick handling areas.
• Traditional disposal methods for solid hatchery waste include land fill, composting, rendering, and incineration.

**a. Power generation**

• The hatchery waste can be automatically fed by conveyor belts into a furnace which is equipped with a rotating shredder unit for chopping and grinding solid waste.
• An incinerator system can be used as a furnace to heat the solid and liquid waste to produce steam.
• The steam can power a turbine generator to produce electricity.

**b. Rendering**

• Simultaneously dries the material and separates the fat from the protein and yields fat and a protein meal should be pathogen free.

**c. Autoclaved and extruded**

• Extruded or autoclaved hatchery waste could be used as livestock feed.

**d. Boiling**

• Hatchery waste should be boiled at 100°C with a pressure of 2.2 kg/cm2 for 15 min; then boiled again at 100°C for 5 hours, followed by boiling at 130°C for 1 h then cooled to ambient temperature.
• Dead embryos could be boiled for 100°C for 30 min, soaked in cold water for 20 min to remove shells, sun dried for 4d and used in poultry feed.

**e. Ensiling**

• The eggs were mixed in a 1:1 ratio with formic and propionic acids for 8 weeks at room temperature.
• The acids act by intervening specifically in the metabolism of the microorganisms involved in spoilage.
• The reduction in the pH creates an environment which is unfavourable for microorganisms. The rapid reduction in the pH diminishes the growth of bacteria which produce butyric acid and ammonia and promotes the growth of lactic acid-producing bacteria.
• The lactic acid is responsible for the low pH necessary for storage of the by-product before being used in animal feed.

**f. Composting**

• Composting is a common method for solid organic waste disposal.
• The decomposition of organic waste is performed by aerobic bacteria, yeasts and fungi. The composting process kills pathogens, converts ammonia nitrogen to organic nitrogen
• The product can be used as a fertilizer.
• Disadvantages of composting are loss of some nutrients including nitrogen.
• Composting with litter eliminates Salmonella
• The hatchery waste can be mixed with wood shavings to reduce the moisture then composted.
• The composter turns manure, litter, sour feed stuffs and carcasses into compost in 4 days with minimal labour and mechanical devices.

**g. Anaerobic digestion systems**

• High efficiency process
• Produces biogas for power generation or heating
• The bio-solids may be used as a high quality fertilizer and generation of electricity
• Anaerobic digestion of organic waste by microbial organisms to produce methane and inorganic products

**3. SLAUGHTER HOUSE WASTE DISPOSAL**

• Rendering is a process of cooking and sterilizing non-edible waste
• Best options for treatment of non-edible wastes by converting waste into meat meal
• Poultry bye-product hydrolyzed feather meal (or PBHFM) or simply Meat Meal.

**Advantages of rendering:**

• Rendering is more effective and profitable
• Converts entire poultry waste into high protein sterilized meat meal
• Prevents environment pollution by disposing of all biological waste
• Meat meal is used for making animal feed

**4. DEAD BIRD DISPOSAL**

**a. Burying**

• Disposal of birds for small farms that cannot construct an incinerator.
• Deep hole may be dug and carcasses buried deeply to prevent worms from carrying infections from the carcass to the surface of the ground
• Deep narrow trench can also be used

**b. Pit disposal**

• Effective and convenient method for disposal of dead birds.
• 150 feet from the poultry houses and water supply
• Flies and insects should not enter the pit
• The pit should be covered with tar paper or plastic
• The pit should be near the post mortem room
• Practical size for pit is about 1.8 m square by 2.4 m deep with drop tube
• Tight fitting lid on the upper end of the tube to prevent the escape of foul odours and the entrance of flies.

**c. Incineration**

• Burning of the carcass
• An incinerator is a furnace used for burning.
• Incineration process uses electricity, firewood or oil
• Electrical or oil-fired incineration is the best available technology
• Rapid destruction of disease-producing organisms, leaving only a small amount of
• Ash which can be distributed on the land
• Smokeless and odourless burning with minimal air pollution

**d. Septic tank disposal**

• Breaking down the carcasses and waste products in an electrically heated septic tank by the action of mesophilic bacteria.
• Heat is applied at 37.8°C and requires 2-3 kwh per day of electricity to maintain this temperature for the two weeks needed for destruction of all but the bones of the carcasses.
• The bacterial action and speed of decomposition can be accelerated by adding lime and hot water at intervals.
• Usually a tank of 2000 litre capacity is required for a flock of 10000 birds.

**e. Composting**

• Composting reduce and transform organic waste into a useful end product called “compost”.

• Alternate layers of litter and paddy straw and dead birds and water
• Finally, the carcasses are covered with a layer of manure.
• Once full, a final cover of litter is placed over the carcasses.
• The temperature of the compost increases rapidly to 60-70°C within 10 days.
• Decomposition starts and kills micro-organisms.
• Temperature decreases after 14-21 days later
• At this point, the material is moved to the secondary bins
• Aerated and allowed for a second rise in temperature.
• The compost material can be safety stored
• 10 m3 of bin space is required for every 1000 kg of carcass.

**f. Rendering**

• Rendering is a heating process that extracts usable ingredients, such as protein meals and fats.
• Rendering converts the inedible results from the slaughtering process into meat meal, bone meal, and feather meal

**The following methods may be followed as pre-treatment or method for rendering**

**1. Daily pickup**

• Daily pickup of poultry carcasses leads to disease transmission.
• Biosecurity should be practiced.
• Central carcass disposal sites should be used for commercial conditions

**2. Freezing**

• Dead birds can be stored on the farm in freezing condition until they can be rendered.
• Freezing reduces or eliminate pollution and improve conditions on the farm

**3. Fermentation**

• Mixes dead birds (which have been ground into 1-inch particles) with a fermentable carbohydrate source, such as sugar, whey, ground corn, or molasses.
• Reduces the pH level so that pathogenic microorganisms are inactivated and the organic materials are preserved.
• Biologically safe, pathogen free safely transported to a rendering plant, recovery of nutrients and recycled into usable foodstuffs or animal feed.

### 4. Acid Preservation

- Propionic, phosphoric, or sulfuric acid is added to carcasses.
- Stored in airtight, plastic containers
- Eliminate the potential for transmitting pathogenic microorganisms

### Advantages of rendering

- Removal of all mortalities from the farm.
- Eliminates environmental pollution
- Nutrient losses, water quality, and recycling for profit increase

# SELF CHECK 1.4-4

## Organic waste collection

Write YES if the statement is correct write NO if it is in correct.

1. Organic waste, or biodegradable waste, is a natural refuse type that comes from plants or animals.
2. Cage layer waste – Excreta collected under the cages, spilled feed and feathers.
3. Composting reduce and transform organic waste into a useful end product called "compost".
4. Freezing reduces or eliminate pollution and improve conditions on the farm
5. Pit disposal is effective and convenient method for disposal of dead birds.

# ANSWER KEY 1.4-4

## Organic waste collection

1. YES
2. YES
3. YES
4. YES
5. YES

## INFORMATION SHEET 1.4-5
## Suitable chicken for harvest selection

**Learning Objective:** After reading this information sheet, you should be able to select Suitable chicken for harvest.

**Introduction:**

The most commonly raised meat chicken is the Cornish cross. These chickens reach harvest weight in as little as 6-8 weeks. Other breeds, like the Rudd Ranger, are bred to be raised on forage and take a little bit longer. Breeds like these may not reach harvest size until they are 12 weeks old. When your chickens have reached harvest size, plan a day to harvest them. It's easier to plan ahead and make sure that you have help, even if you're only harvesting a handful of chickens.

Remove feed from the chicken's coop the evening before you plan on harvesting them. Chickens will consume feed and fill their crop up. A full crop makes the cleaning portion of harvesting more difficult. It's much easier to clean up a chicken that had an empty crop than one with a full crop.

You'll need kill cones, a sharp boning knife, a pot of scalding water and shrink wrap bags made for chickens. If you don't have kill cones, you can clean up old milk jugs and cut the bottoms out of them. Hang them upside down on a tree in place of a kill cone.

Put the chickens head first into the kill cone and pull the head gently through the end. On the side of the chicken's neck, locate the vein. Use the sharp boning knife to sever the vein. Be careful not to cut the windpipe. The goal is to bleed the chicken out completely. If you cut the windpipe, the chicken may die before all of the blood has left the chicken. It will take a few minutes for the chicken to die. Once the chicken has stopped bleeding and moving, it's ready for the scalding water.

Scalding water is used to loosen the feathers on the chicken so that plucking is easier. We use a large crawfish pot or turkey cooker over a propane flame as a scalding pot. Fill it with water before you start killing chickens and get it heating up. The water should be steaming and just before boiling, around 190 degrees. Grab the chicken by the feet and completely submerge it into the water for 3-5 seconds. Pull it back out of the water and try to remove some of the feathers. If they slip right out, start plucking it. If the feathers still resist, then dip it back into the water again.

When all of the feathers are removed, you can cut off the head and feet. If you want to save the feet, dip them into the scalding tank and rub off the yellow layer of skin. Carefully cut around the cloaca, making a hole large enough to remove the innards. Be careful not to cut any of the intestines in the process. Take the boning knife and cut around the esophagus and trachea to loosen them. When you pull the intestines and other innards out, they should come out with it.

Clean up the carcass and place them into the shrink wrap bags. You can seal the bags now or later. Shrink wrap bags are dipped into hot water and shrink up around the chicken. Put the chicken in the refrigerator for 24-48 hours. This allows the chicken's muscles to relax and helps it not be tough. After time in the fridge, they can go into the freezer.

If you're looking for reasons to raise dual purpose breeds that both lay eggs and provide meat

## Meat bird types

Butchering an old laying hen will produce a stewing chicken that can be used for a tasty soup or casserole, as well as for making chicken stock for general use. Meat chicken breeds raised exclusively for butchering are best if you want to fry or roast the meat

## SELF CHECK 1.4-5
## Suitable chicken for harvest selection

Write YES if the statement is correct write NO if it is in correct.

1. When your chickens have reached harvest size, plan a day to harvest them.
2. Remove feed from the chicken's coop the evening before you plan on harvesting them
3. . A full crop makes the cleaning portion of harvesting more difficult
4. . A full crop makes the cleaning portion of harvesting more difficult
5. When all of the feathers are removed, you can cut off the head and feet.

## ANSWER KEY 1.4-5
## Suitable chicken for harvest selection

1. YES
2. YES
3. YES
4. YES
5. YES

**INFORMATION SHEET 1.4-6**
**Production record**

**Learning Objective:** After reading this information sheet, you should be able to appreciate the importance of production record.

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**Introduction:**

Put in mind that for you to succeed, income and expenses have to be factored in so that you can know if your business venture is sustainable or not.
Records usually show the weaknesses as well as strengths of your agribusiness.

As a business owner, you will be able to tell where your business has been and the direction it is heading.

**Why Record Keeping is Vital.**

- For you to make critical financial decisions as well as create a budget, you will need to analyze your records.
- For you to make feeding decisions especially on the number of feed rations, types of feed rations and the effectiveness of the specific feed ration you will need to look at your records.
- A record will help you identify mistakes that can be avoided in the future.
- You should keep records on the productivity of your chicken such as kilos of meat, reproduction, eggs that will be used when culling your flock.
- When you need to breed your birds, you will need to analyze your records to be able to identify the most productive ones.
- Records will help you study the production performance and check whether it meets the prescribed standards
- Most banks will look at your records before giving you any loan.

**Methods of Record Keeping.**

**Hand recording systems.**

**Advantages of hand recording system.**

- Easy to start.
- Low initial out of pocket expense.
- Only requires paper and pencil.

**Computer recording system.**

**Advantages of a computer recording system**

- Easier to create analysis.
- Accurate and faster.
- Tax deductible as an expense.

**Chicken farming records classification.**
Records should be simple, easy to understand, with no repetition and have all the necessary information. Chicken records can be divided into three categories. They include;

**Management records.**

Management records usually include data that is related to management issues. They include number, date, and type of vaccination or medication administered, type and amount of feed given, death or loss of chicken, date of chick placement as well as workers shift.

In layers, it should include management practices such as beak trimming, de-worming dates, culled birds, disease incidence and the steps taken.
Management records will help you when assessing the levels of production of your chicken. You can promptly go back in time and know where things started going wrong. You will also be able to know if you are on the right track.

**Financial records.**

Financial records are supposed to contain all the financial transactions in your chicken farming ventures. You, therefore, should be very keen when entering records of huge expenditures such as those of feeds and purchase of day-old chicks. Ensure that you also keep any financial records of any items sold or bought including supplies, veterinary costs, and equipment.

Remember not to overlook any production cost. It usually adds up to a lot of money in the end. For example, in broiler production, you can easily ignore the cost of slaughtering and transporting the meat to the market. In the end, your financial records will not give you an accurate production cost.

**Production records.**

Production records are vital in assessing productivity. For example, in broilers, the daily or weekly weight gain indicates productivity. You can use that to compare the records to the standard chart. You can then make changes in the feeding program if you see the need to.
Many farmers usually weigh their chicken right before slaughter. However, the best way is to weigh weekly to identify any negative deviations. You will be in a position to minimize losses because you can make changes as soon as you notice a deficiency.

In layers, your record should include dates, eggs produced, opening balance, and eggs sold as well as the closing balance of the eggs. Ensure to include daily day to day sales of your eggs. Never forget to maintain a record of the number and sale price of breakable saleable eggs as well as those of the pullet eggs.

It is time you embrace record keeping for your chicken farming. Whatever methods you use in record keeping, whether a computer or hand you should ensure that you maintain and categorized correctly. It better to identify problems right after it happens

## SELF CHECK 1.4-6
## Production record

Write YES if the statement is correct write NO if it is in correct.

1. Records usually show the weaknesses as well as strengths of your agribusiness
2. Management records usually include data that is related to management issues.
3. Production records are vital in assessing productivity. For example, in broilers, the daily or weekly weight gain indicates productivity
4. A record will help you identify mistakes that can be avoided in the future
5. Records will help you study the production performance and check whether it meets the prescribed standards

**ANSWER KEY 1.4-6**
**Production record**

1. YES
2. YES
3. YES
4. YES
5. YES